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Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation

Master Thesis at Ulm University

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1 Introduction

1.1 Kontext

The General Data Protection Regulation (GDPR) requires organizations to maintain a Record of Processing Activities (ROPA), a formal document that outlines how personal data is collected, processed, stored, and shared. Creating and maintaining these ROPA documents is often a resource-intensive process, especially for small and medium-sized organizations that lack specialized legal or data-protection expertise. Previous research developed an application that leverages Large Language Models (LLMs) to support users in generating GDPR-compliant ROPA documentation. The application provides three distinct interaction modes, each offering different degrees of assistance from the LLM, from basic text suggestions to highly automated drafting of ROPA entries. This system aims to reduce the effort needed to create legally compliant documentation while maintaining transparency and user control. Although the application provides a functional framework for assisted ROPA generation, its effectiveness, usability, and impact on documentation quality have not yet been systematically evaluated. To ensure that such a system can be confidently integrated into practical compliance workflows, both qualitative and quantitative assessments are needed. These evaluations must determine whether users can efficiently and correctly create ROPA documents, how the varying levels of LLM support influence performance and trust, and whether the system produces outputs aligned with GDPR requirements.

1.2 Research Problem and Research Questions

The goal of this thesis is to conduct a comprehensive evaluation of the existing LLM-supported ROPA generation application. The work will focus on assessing how the different interaction modes affect user experience, efficiency, output quality, and perceived reliability.

RQ1:

- How do varying degrees of LLM support influence Novice users during ROPA document creation?
- How do novice users perceive the usability, efficiency, and satisfaction of the three interaction modes (Form, Ask, Chat)?
- How does cognitive load differ across the three modes for users with no prior GDPR knowledge?
- Which mode do novice users find most intuitive and prefer, and why?
- How confident do users feel about the quality of their generated documents in each mode? (This is now crucial - perception vs. reality)

RQ2:

- How do varying degrees of LLM support influence Document quality when used by novice users?
- How does document quality differ across the three generation modes based on semantic similarity to an expert-created reference (BERTScore)?
- Compare documents using an LLM
- Are all fields present? (Does my pipeline work)
- Is there a discrepancy between user confidence (RQ1) and document quality (RQ2)?

1.3 3. Teil: Objectives und Contributions

Thesis Organization

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2 Background

2.1 GDPR

The General Data Protection Regulation (GDPR), implemented on May 25, 2018, represents the most stringent and comprehensive privacy and security framework in the global digital landscape [5]. While originated and enacted by the European Union, the regulation maintains an extraterritorial scope, imposing legal obligations on any organization—regardless of its physical location—that collects or processes data belonging to individuals within the EU. This landmark legislation is rooted in the 1950 European Convention on Human Rights, which established privacy as a fundamental right. It serves as a necessary evolution of the 1995 European Data Protection Directive, modernizing legal protections to address the rapid proliferation of cloud services, social media, and large-scale data breaches that have characterized the early 21st century. [5, 4]

Under the GDPR, the definitions of roles and data types are expanded to capture the complexity of modern informatics. Personal data is broadly defined as any information relating to an identifiable individual, encompassing everything from names and email addresses to biometric data, location information, and web cookies. The regulation distinguishes between the data subject (the individual), the data controller (the entity determining the "why" and "how" of processing), and the data processor (third parties handling data on behalf of a controller). To ensure ethical handling, all processing must adhere to seven core principles: lawfulness, fairness and transparency; purpose limitation; data minimization; accuracy; storage limitation; integrity and confidentiality; and accountability. This last principle is particularly vital, as it shifts the burden of proof to the organization, requiring documented evidence of compliance. [5]

The regulation mandates that processing is only legal if justified by one of six specific bases, most notably through consent, which must be freely given, specific, and unambiguous. Other justifications include contractual necessity, legal obligations, or a legitimate interest that does not override the subject's rights. Furthermore, the GDPR introduces the concept of data protection by design and by default, requiring that privacy considerations be integrated into the initial development phases of any product or service. Large-scale processors or public authorities are often required to appoint a Data Protection Officer (DPO) to oversee these internal processes and serve as a liaison with regulatory bodies. [5]

For individuals, the GDPR grants a robust suite of privacy rights, including the right to be informed, the right of access, the right to rectification, and the right to erasure (also known as the "right to be forgotten"). Subjects also possess the right to data portability and protections against automated decision-making and profiling. To enforce these standards, the regulation possesses significant "teeth"; authorities can levy harsh penalties for non-compliance. Consequently, the GDPR has transformed data privacy from a secondary IT concern into a primary strategic and legal priority for enterprises worldwide. [5].

2.2 Records of Processing Activities under Article 30 GDPR

Article 30 of the General Data Protection Regulation (GDPR) establishes the obligation for controllers and, where applicable, processors to maintain a Record of Processing Activities (RoPA). The RoPA functions as a core accountability instrument within the GDPR framework, operationalizing the principle that organizations must not only comply with data protection requirements but also be able to demonstrate such compliance to supervisory authorities. Its primary purpose is to provide a structured, transparent, and up-to-date overview of how personal data are processed within an organization, thereby enabling effective oversight, risk assessment, and regulatory control. [4, 14]

From a regulatory perspective, the RoPA serves several interrelated objectives. First, it enhances organizational awareness of data processing operations by requiring a systematic mapping of purposes, data flows, and safeguards. Second, it supports data protection by design and by default, as organizations must explicitly consider data categories, retention periods, and security measures. Third, it enables supervisory authorities to efficiently assess compliance, particularly in the context of audits, investigations, or following personal data breaches. As such, the RoPA is not merely a formal record, but a living compliance document that reflects actual processing practices. [4]

Article 30 distinguishes between the obligations of data controllers and those of data processors. Controllers are required to document processing activities carried out under their responsibility, while processors must maintain records of processing performed on behalf of controllers. These records must be maintained in writing, including in electronic form, and must be made available to the competent supervisory authority upon request. Although the GDPR provides a limited exemption for organizations employing fewer than 250 persons, this exemption does not apply where processing is non-occasional, involves special categories of personal data, concerns criminal convictions, or is likely to result in risks to the rights and freedoms of data subjects. Consequently, in practice, many small and medium-sized organizations remain subject to the RoPA requirement. [4]

The content of a RoPA is explicitly defined by Article 30 GDPR and is intended to be concise, structured, and easily comprehensible. Excessive or irrelevant information should be avoided, as clarity and usability are essential for both internal governance and external supervision. At a minimum, a RoPA should include the following elements.

- Name and contact details of the data controller (and, where applicable, joint controllers, representatives, and the data protection officer).
- Purposes of the processing activities.
- Categories of data subjects.
- Categories of personal data processed.
- Categories of recipients, including recipients in third countries or international organizations.
- Details of transfers of personal data to third countries or international organizations, including safeguards where applicable.

- Time limits for the erasure of different categories of personal data.
- A general description of the technical and organizational security measures implemented pursuant to Article 32 GDPR.

Where an organization acts as a data processor, the RoPA must additionally reflect the processing carried out on behalf of each controller, including the identity of those controllers, the categories of processing involved, relevant international data transfers, and applicable security measures. In both cases, the RoPA should be sufficiently detailed to provide a faithful representation of processing activities, while remaining accessible to non-technical stakeholders. [4]

The GDPR does not prescribe a fixed update cycle for RoPAs; instead, it requires that records remain accurate and up to date. Consequently, any substantive change in processing purposes, data categories, recipients, or safeguards should trigger a review and update of the RoPA. Many organizations delegate the responsibility for maintaining the RoPA to a data protection officer, where one is appointed, to ensure consistency, avoid duplication of effort, and reduce the risk of omissions. [4, 11]

Finally, Article 30 emphasizes the external relevance of RoPA. Organizations must be prepared to present their records to supervisory authorities upon request, and such requests may be accompanied by demands for supplementary documentation, including privacy notices, consent records, or contractual arrangements. In this sense, RoPA operates as a foundational compliance artifact, anchoring broader data protection governance and demonstrating an organization's commitment to accountability under the GDPR. [4, 11].

2.3 Related Work

Prior work relevant to this thesis can be grouped along four lenses that mirror the design choices of the present study. The first concerns *domain*: research that targets GDPR compliance specifically and the automated production of regulatory artefacts such as ROPA. The second concerns *task*: architectures for generating structured, template-bound legal documents in adjacent jurisdictions. The third concerns *evaluation*: how the legal-AI literature measures the quality of generated documents, increasingly through a combination of surface-overlap metrics, semantic-similarity metrics, and either expert or LLM-based judgement. The fourth concerns the *user*: how legal AI is presented to non-expert audiences and whether it is accessible to people without prior legal training. Reviewing the literature through these four lenses makes clear what has been studied, what has not, and where the present thesis sits.

2.3.1 GDPR Compliance Automation

The closest prior work to this thesis targets the automation of GDPR compliance documentation directly. von Schwerin et al. [14] address the generation of data categories (e.g., contact information, payment data) for Records of Processing Activities, observing that existing legal benchmarks such as LawBench or Legal-Bench primarily evaluate static knowledge recall rather than the ability to produce documents that adhere to the precise formatting and terminology required by the GDPR. Their study compares four open-source models (Qwen2-7B, Meta-Llama-3-8B, zephyr-7b-beta, and orca_mini_v7_7b) against GPT-4 across four experiments—basic performance, model-specific prompting, few-shot prompting, and Retrieval-Augmented Generation (RAG)—and assesses outputs using a weighted Model Utility Score covering latency, structure and style, grammar, validity, and contextual understanding. The finding that open-source models, particularly Qwen2-7B, can match or exceed GPT-4 on structured ROPA outputs validates the use of locally deployable models for privacy-sensitive compliance work, and is the empirical foundation on which ropagen builds. The study also documents persistent risks—ambiguous legal language, jurisdictional variation, hallucination, and the resulting need for a "human-in-the-loop"—that motivate examining how real users behave when asked to produce ROPA documents with such tools.

Abualhaija et al. [1] approach the GDPR from a complementary angle, extracting technical software requirements for specific data subject rights (right of access, right to portability) from the regulation and supplementary legal sources. Their XTRAREG pipeline uses GPT-3.5 and GPT-4o together with RAG and prompting strategies (Zero-Shot Learning, Chain-of-Thought) to generate requirements, validated against a manual baseline of 108 expert-extracted reference requirements. The work confirms two patterns that recur throughout this thesis: RAG meaningfully improves both lexical and semantic alignment with expert references, and high

BERTScores can coexist with low BLEU when the model captures the meaning of a requirement without matching its wording. Their reported failure modes—incomplete coverage, perspective drift (writing as the data subject rather than as the system), and citation of non-binding recitals—further reinforce the case for evaluating LLM-generated GDPR artefacts against a fixed reference rather than trusting model output at face value. Together, these two works establish GDPR documentation as a viable LLM target while leaving the user-side question unanswered: how do non-experts fare when asked to produce such documents themselves? ropagen, and this thesis, extends this line of work by testing the user-facing realisation of an open-source GDPR pipeline rather than the model performance in isolation.

2.3.2 Structured Legal-Document Generation

A second body of work targets the broader problem of producing structurally coherent legal documents from limited user input, exploring the architectural strategies—retrieval, wrappers, fine-tuning—that constrain LLMs to follow rigid legal templates. Nigam et al. [10] address private legal-document drafting in the Indian system through their Model-Agnostic Wrapper (MAW), a two-phase framework in which the model first generates a structured list of section titles (which the user can manually refine) and then iteratively fills each section, using a vector database (ChromaDB) to summarise prior sections and maintain long-form coherence. The system is exposed through a Human-in-the-Loop interface and evaluated on the Vidhik-Dastaavej dataset of 489 anonymised documents using ROUGE, BLEU, METEOR, BERTScore, an automatic G-Eval LLM judge, and a Likert-scale expert assessment. The headline finding—that direct fine-tuning on small datasets degrades quality, while a structured wrapper around a smaller open-source model can match GPT-4o—argues for orchestration over scale.

Lin and Cheng [9] take the opposite architectural route, fine-tuning an open-source BLOOM-560M model locally on 74,823 unlabelled Traditional Chinese fraud verdicts (Taiwan Judicial Yuan, 2011–2021) without Chinese word segmentation. Their structural-correctness metric defines a fixed order of legal constitutive elements—Subject of Crime, Subjective Legal Element, Act, Victim, Causation, Result of Hazard—and checks whether the generated draft follows that order, supplemented by perplexity and training/validation loss. The work demonstrates that domain-specific structured drafting is feasible without cloud infrastructure on consumer-grade hardware, addressing the same privacy concerns that motivate locally deployed GDPR tooling. Li et al. [7] push the structural framing further with CaseGen, a benchmark that decomposes Chinese case-document drafting into four sequential stages—defence statements, trial facts, legal reasoning, and judgment results—over 500 expert-annotated cases. Their finding that general-domain LLMs (GPT-4o, Qwen2.5) outperform legal-specific models like ChatLaw or LexiLaw, while still scoring below 6 out of 10 overall, suggests that careful staging of the generation task matters at least as much as legal-domain pretraining. Together, these three works frame structured legal-document generation as primarily an architectural problem and explore wrappers, fine-tuning, and multi-stage decomposition as alternative solutions. ropagen takes a related but distinct stance: rather than comparing architectures, it fixes the architecture and varies the *interaction modality* (Form, Ask, Chat), asking what the user-facing surface contributes once the underlying model is held constant.

2.3.3 Evaluation Methodologies for Legal AI

A third strand of the literature concerns how generated legal documents are assessed, and a clear methodological pattern is emerging: dual evaluation that combines automated NLP metrics with either expert annotation or LLM-as-a-Judge scoring. Su et al. [15] formalise this for Chinese judgment-document generation through JuDGE, a benchmark of 2,505 fact-judgment pairs supplemented by over 55,000 statutory articles and 103,000 past judgments. The framework evaluates outputs along four dimensions—Penalty Accuracy (normalised absolute difference on prison terms and fines), Convicting Accuracy (Recall, Precision, F1 over charges), Referencing Accuracy (correctness of statutory citations), and Documenting Similarity (METEOR and BERTScore against ground truth)—explicitly arguing that lexical metrics alone cannot capture the legal reasoning that distinguishes a sound judgment from a plausible-sounding one. Their results across direct generation, in-context learning, supervised fine-tuning, and Multi-source RAG show that even the strongest pipelines remain visibly short of professional ground truth, and that semantic-similarity metrics behave as proxies rather than ceilings.

Vayadande et al. [17], viewed from the methodological angle, illustrate the second template: comparative quantitative benchmarking against existing market tools

(Kira Systems, Evisort) using accuracy percentages, time-efficiency in hours, and document-processing speed, supplemented by a debugging phase in which legal experts contribute "balanced comments" that refine the system. The reported 93–97 % accuracy and reduction of document time from 20 to 5 hours sit alongside a candid acknowledgement of remaining failures on judicial nuance and on input-quality dependencies, foreshadowing the same tension between aggregate metrics and qualitative caveats that emerges in this thesis. Read together, JuDGE and Vayadande et al. converge on a shared template: surface-overlap metrics (BLEU, ROUGE, METEOR), semantic-similarity metrics (BERTScore, sentence embeddings), and a complementary qualitative judgement layer—human expert or LLM judge—that catches what the lexical and semantic metrics miss. This is the methodological template adopted in the present study for evaluating ropagen's three modes against a single expert reference document.

2.3.4 User-Facing Legal AI Systems

The fourth lens concerns how legal AI is presented to non-expert users, where prior work is markedly thinner. Surya et al. [16] target the accessibility gap directly with an AI-powered interactive legal chatbot for the Indian Department of Justice, aiming to democratise legal information for citizens facing complex jargon and the cost of professional counsel. Their three-tier architecture—a JavaScript single-page frontend, a Python Flask orchestrator handling translation, and an LLM service running LLaMA 3.1 via Ollama—supports real-time legal queries, document summarisation (e.g., First Information Reports), and multilingual interaction by translating Hindi queries to English for the LLM and back. The system also surfaces citation links to government portals, addressing the trust problem that arises whenever non-experts consume LLM output on a high-stakes topic. Notably, however, the paper evaluates these features through illustrative use cases rather than a controlled user study, leaving open how non-expert users actually interact with such a system in practice.

Vayadande et al. [17], viewed from the UX angle rather than the methodological one, fit the same pattern: their assistant is explicitly framed as a tool to make legal practice "more frictionless" for users, integrating ChatGPT 3.5 with OpenAI Embeddings, LangChain, FAISS vector stores, PyPDF, Amazon Textract, and Named Entity Recognition with sentiment analysis to support generation, comprehension, and abnormality detection. Yet the system is again validated through accuracy and efficiency comparisons rather than through evidence of how non-expert end-users navigate the interface, what they understand of the output, or how confident they feel in what the system produces. Across both works, the user-facing legal AI literature establishes that accessibility, multilingual support, and citation transparency matter, but stops short of empirically measuring how well the resulting systems serve the non-experts they target.

2.3.5 Synthesis

Read across these four lenses, the literature establishes that LLM-based legal-document generation is technically feasible, that architectural choices (wrappers, fine-tuning, multi-stage decomposition, RAG) measurably improve output, and that dual evaluation combining NLP metrics with expert or LLM-as-a-Judge scoring is becoming the de facto standard. What remains absent is the intersection of three concerns. No prior work simultaneously (a) targets GDPR-specific structured output, (b) exposes the system to *novice users* without legal training, and (c) measures the gap between user confidence and actual document quality under varying levels of LLM support. This thesis fills that gap by holding the underlying ropagen pipeline constant, varying the interaction modality across Form, Ask, and Chat, and applying the dual-evaluation template from the legal-AI literature to a within-subjects user study against a single shared scenario.

3 Methodology

3.1 User Study

Study Design

The study followed a within-subjects experimental design with counterbalanced ordering, using a Balanced Latin Square. The recruited sample consisted of $N = 73$ university students, all novice users with no prior GDPR experience, and each session lasted approximately 90 minutes per participant. Counterbalancing was implemented through a Balanced Latin Square with six groups, reflecting the three AI modes tested in randomised sequences; participants were randomly assigned to one of the six groups, and each completed all three modes in a predetermined counterbalanced order. This design controls for order effects, learning effects, and fatigue.

Task Scenario

A single standardised scenario was used across all participants and modes to ensure consistency. The scenario described a relatable university-based data processing activity covering all GDPR Article 30 fields, ensuring comparability across conditions while remaining accessible to participants without legal expertise. The underlying LLM was Mistral Large 3 (model identifier `mistralai/mistral-large-2512`), held constant across all modes and routed via OpenRouter.

Participants

Recruitment yielded $N = 73$ university students. Participants were screened for minimal to no GDPR knowledge (expected and required), prior AI tool usage experience (documented), general technology comfort level (assessed), and the absence of any professional legal or compliance background. As part of onboarding, participants received a brief standardised introduction to GDPR and ROPA concepts before beginning the tasks, and were shown an example of a high-quality ROPA document as reference material prior to starting.

Data Availability and Sample Sizes

Due to technical data loss and incomplete submissions during data collection, the effective sample sizes differ across analysis streams. The canonical sample sizes used throughout this thesis are summarised below; all counts are derived from the raw data files committed to the repository and documented in full in `python/output/sample_size`

- **Survey, recruited N = 73 / valid N = 70** — 73 participants completed the study; 70 retained after dropping cases with invalid consent or quality flags. Per-mode sample size: **n = 69** for each of Form, Ask, and Chat (one participant has a valid Form response but no Ask/Chat). The matched-triple survey subset (participants with valid responses for all three modes) is **n = 69**; this is used for within-subjects Friedman and Wilcoxon signed-rank tests on SUS, R-TLX, confidence items, and the perceived AI support composite.
- **Documents, 113 generated ROPAs** — distributed across modes as 37 (Form), 37 (Ask), and 39 (Chat); used for all NLP metric and LLM-as-a-judge analyses at the document level. The 113 documents represent **47 unique participants**; **28 participants** have at least one document in each of the three modes (loose match), and **26 participants** have exactly one document per mode (strict matched triple). The strict **n = 26** set is used for all within-subjects tests on document-quality measures (Friedman, Wilcoxon signed-rank); the unbalanced full sample (n = 113) is used only for between-groups sensitivity checks (Kruskal–Wallis, Mann–Whitney U).
- **Survey × NLP join, 106 rows** — inner join on participant and mode yields

106 rows (Form 34, Ask 34, Chat 38) across **44 unique participants**. Per-mode joined samples (n = 34 / 34 / 38) are used for per-mode Spearman correlations between confidence and quality, which do not require complete triples.

- **Triangulated sample, n = 27** — participants with survey × NLP coverage in all three modes. This is the canonical N for confidence–quality correlation analyses that require within-subjects matching across all three modes simultaneously.

The reconciliation between these canonical Ns and earlier reporting is as follows: 47 corresponds to unique participants with at least one judge-evaluated document, 28 to participants with at least one document per mode (loose match), and 44 to unique participants in the survey × NLP join. The canonical figures used in the inferential analyses are **n = 26** for paired NLP/judge tests and **n = 27** for triangulated correlations across all three modes.

Descriptive statistics (means, medians, standard deviations) are computed on the full available N per mode for each analysis stream. Within-subjects inferential tests are restricted to the matched-triple subsets specified above. Where the unbalanced full sample makes matching impossible, between-groups sensitivity checks are reported alongside the matched analyses (see *Statistical Analysis*, below).

Data Collection

Quantitative measures were collected after each mode. The System Usability Scale (SUS) was administered as ten standardised questions on a 5-point Likert scale, and the Raw Task Load Index (R-TLX) captured six cognitive load dimensions, also on a 5-point Likert scale. The full text of the 10 SUS items and the 6 NASA R-TLX subscale descriptions is reproduced in Appendix B. In addition, participants completed a six-item Perceived AI Support instrument on a 5-point Likert scale:

1. *Perceived Usefulness:* "The LLM was helpful in creating this document."
2. *Efficiency:* "This mode let me complete the ROPA faster than I could on my own."
3. *Transparency:* "I understood why the AI suggested certain GDPR fields."

4. *Reliance Comfort:* “I felt comfortable relying on the AI’s suggested categories.”
5. *Task-Specific Confidence (primary confidence measure):* “I am confident the AI identified the appropriate legal basis.”
6. *Intention to Use:* “I would use this mode again for similar work.”

Item 5 is treated as the **primary confidence measure** throughout the analysis: it is the only item that asks participants to judge the substantive correctness of the AI’s output (the legal basis under Article 6 GDPR), and is therefore directly comparable to the document-quality signals from the NLP and judge pipelines. The remaining five items (usefulness, efficiency, transparency, reliance comfort, intention to use) capture distinct facets of *perceived AI support* rather than confidence in output correctness, and are reported as a secondary composite. Free-text qualitative feedback was also collected after each mode, and total session duration was recorded as a single value per participant; per-mode timing was not recorded. After completing all three conditions, participants ranked all three modes by overall preference in a post-experiment ranking task.

Quality Assessment

Following the dual-evaluation template adopted by Su et al. [15] and Vayadande et al. [17], both surface-form and semantic measures are reported alongside an LLM-as-Judge rubric.

NLP Metrics

All participant-generated documents (73 participants $\times$ 3 modes = 219 documents) are compared against a single reference document using a suite of automated NLP metrics. The reference document was created by the study author using the Form mode of ropagen, applying the same standardised scenario presented to participants. This document is treated as the authoritative baseline — a root-of-trust representing a careful, complete application of the most structured interaction mode. It should be noted that this introduces an inherent framing effect: NLP metrics measure similarity to Form-style output, which may disadvantage the more conversational outputs of Ask and Chat modes independently of their substantive quality.

- **BLEU** (Bilingual Evaluation Understudy): n-gram precision with brevity penalty
- **ROUGE-1, ROUGE-2, ROUGE-L**: unigram overlap, bigram overlap, and longest common subsequence F1
- **METEOR**: precision–recall balance with synonym matching
- **BERTScore** (Precision, Recall, F1): token-level semantic similarity using ModernBERT embeddings
- **SBERT Cosine Similarity**: document-level semantic similarity using sentence embeddings

Metrics are computed at both the document level (aggregated across all Article 30 sections) and the chapter level (per individual GDPR section). Results are aggregated by mode to establish comparative quality baselines.

Metric Definitions

The five NLP metrics summarised above are defined formally below. Each is computed against the single expert-authored reference document; interpretive framing of the resulting values is deferred to the Evaluation chapter.

BLEU rewards exact word-sequence overlap with the reference: a high BLEU score means the candidate reuses the reference's wording almost verbatim, while a low BLEU score can equally indicate a legitimate paraphrase or genuinely missing content – the metric does not distinguish the two. It is therefore most informative when read alongside a recall-aware or semantic measure.

BLEU [12] computes the geometric mean of n-gram precisions p_n between candidate and reference, multiplied by a brevity penalty (BP) that discourages outputs much shorter than the reference. With uniform weights $w_n = 1/N$, candidate length c, and closest reference length r:

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \quad \mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r. \end{cases} \tag{3.1}$$

This work uses $N = 4$ (matching unigrams through 4-grams).

ROUGE measures how much of the reference's vocabulary, bigrams, or longest common subsequence the candidate covers. A high ROUGE score signals strong lexical recall against the reference and tolerates word re-ordering more readily than BLEU; a low score indicates that key reference content is absent from the candidate.

ROUGE [8] reports the F1 of n-gram or longest-common-subsequence (LCS) overlap between candidate X and reference Y. ROUGE-1 counts unigrams, ROUGE-2 counts bigrams, and ROUGE-L counts the LCS. For ROUGE-N:

$$P_N = \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(X)|}, \quad R_N = \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(Y)|}, \quad F1_N = \frac{2P_N R_N}{P_N + R_N}. \tag{3.2}$$

The same precision–recall–F1 form applies to ROUGE-2:

$$F1_{\text{ROUGE-2}} = \frac{2P_2 R_2}{P_2 + R_2}. \tag{3.3}$$

For ROUGE-L, n-gram counts are replaced by the length of $\mathrm{LCS}(X, Y)$:

$$P_L = \frac{|\mathrm{LCS}(X,Y)|}{|X|}, \quad R_L = \frac{|\mathrm{LCS}(X,Y)|}{|Y|}, \quad F1_L = \frac{2P_L R_L}{P_L + R_L}. \tag{3.4}$$

The reported ROUGE values are the F1 scores in each case.

METEOR balances precision and recall, credits synonym and stem matches, and penalises fragmented overlap where shared words appear in scattered positions. It is stricter on recall than ROUGE but more permissive than BLEU when the candidate paraphrases the reference using related vocabulary.

METEOR [2] is the weighted harmonic mean of unigram precision P and recall R (with recall up-weighted by α), multiplied by a fragmentation penalty that grows with the number of disjoint matching chunks ch relative to the number of matched unigrams m:

$$\mathrm{METEOR} = \underbrace{\frac{P \cdot R}{\alpha P + (1-\alpha) R}}_{F_{\mathrm{mean}}} \cdot \underbrace{\left(1 - \gamma\left(\frac{\mathrm{ch}}{m}\right)^{\beta}\right)}_{\text{penalty}}. \tag{3.5}$$

Standard values $\alpha = 0.9$, $\beta = 3$, $\gamma = 0.5$ are used; matching is performed against exact, stemmed, and synonym/paraphrase matches via WordNet.

BERTScore measures similarity between the contextual embeddings of candidate and reference tokens rather than their surface forms. *Precision* captures how many candidate tokens find a close meaning-match in the reference and therefore penalises spurious or invented content; *recall* captures how many reference tokens are covered and therefore penalises omission; *F1* is the harmonic mean and requires both to be reasonably high. A high F1 paired with a low BLEU typically indicates a successful paraphrase that preserves meaning without reproducing wording.

BERTScore [19] replaces surface-form n-gram matching with cosine similarities between contextual token embeddings produced by a pretrained transformer (here, ModernBERT [18] via the `bert_score` library). With L2-normalised embeddings $\mathbf{x}_i$ for reference tokens and $\hat{\mathbf{x}}_j$ for candidate tokens:

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.6}$$

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.7}$$

$$F1_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}. \tag{3.8}$$

SBERT compares two documents through a single whole-document embedding, so two ROPAs that cover the same topics with very different wording will still score close to 1. It is the most lenient of the metrics reported here and is used as a coarse topical-coverage signal rather than a fine-grained quality measure.

SBERT [13] produces a single fixed-length embedding per document by mean-pooling token embeddings from a transformer (here, ModernBERT). Document similarity is the cosine of the angle between two document embeddings $\mathbf{e}(d_1)$ and $\mathbf{e}(d_2)$:

$$\mathrm{SBERT}(d_1, d_2) = \frac{\mathbf{e}(d_1)^\top \mathbf{e}(d_2)}{\|\mathbf{e}(d_1)\| \, \|\mathbf{e}(d_2)\|}. \tag{3.9}$$

The precision/recall asymmetry of these metrics maps onto distinct failure modes of the generated documents. Low precision combined with high recall typically reflects a verbose candidate that covers the reference but adds material not present in it; high precision combined with low recall reflects a conservative candidate that omits required content but does not invent. A high BERTScore F1 paired with a low

BLEU is the signature of a successful paraphrase – the document expresses the same content in different words – whereas the rare opposite pattern of a high BLEU paired with a low BERTScore F1 indicates mechanical surface copying without a corresponding semantic match. SBERT, operating at the document level, mostly responds to whether the ROPA addresses the same broad topics as the reference and is used as a sanity check rather than a discriminating metric. These interpretive heuristics are applied throughout the Evaluation chapter.

LLM-as-a-Judge Evaluation

Documents are additionally evaluated using an LLM-as-a-judge approach, in which **Gemini 3.1 Pro Preview** (model identifier `google/gemini-3.1-pro-preview`, accessed via OpenRouter at temperature 0) serves as an automated evaluator assessing GDPR-specific quality. The judge is positioned as a *supplementary* evaluator: it is a single-pass, single-model assessment without inter-rater replication or legal-expert validation, and its scores are reported alongside, not in place of, the NLP metrics. The reference-based NLP metrics remain the primary RQ2 evidence; the judge complements them by providing a reference-free signal that captures GDPR-specific quality properties (legal correctness, hallucination, scenario faithfulness) which similarity-based metrics cannot directly observe. The judge evaluates each document across five dimensions on a 1–5 scale:

1. **Completeness:** Coverage of all required Article 30(1) fields (controller identity, DPO contact, purposes, legal basis, data categories, recipients, retention periods, technical and organisational measures)
2. **Scenario Faithfulness:** Adherence to the specific details of the study scenario rather than generic or hallucinated content
3. **Legal Correctness:** Appropriate use of GDPR terminology and accurate citation of relevant articles
4. **Hallucination (inverse):** Absence of invented facts not present in the scenario (scored inversely: 5 = no hallucinations)
5. **Overall:** Holistic assessment of the document's usability as a real-world ROPA record

The judge evaluation is applied to all 113 retained documents. The full evaluation rubric and system prompt used for the judge are provided in Appendix C.1. The interpretive narrative summarising judge scores in the accompanying analysis artefact (`judge_interpretation.md`) was drafted by Claude Sonnet 4.6 from the raw per-document scores and rationales as a working aid; the interpretation reported in this thesis is authored by the researcher directly from the same raw judge scores.

Confidence–Quality Gap Analysis

To investigate whether perceived confidence in the AI-generated output corresponds to actual document quality, self-reported confidence is correlated with both NLP metric scores and LLM-as-a-judge scores. The **primary confidence measure** is the single Likert item on legal-basis confidence (item 5 of the perceived AI support instrument: "I am confident the AI identified the appropriate legal basis"), chosen because it is the only survey item that asks participants to judge the substantive correctness of the AI's output. The 6-item Perceived AI Support composite is reported as a **secondary** analysis to capture broader attitudinal facets (usefulness, efficiency, transparency, reliance comfort, intention to use). Both measures are correlated against three signals: confidence against NLP metrics (reference-based lexical and semantic similarity); confidence against LLM judge scores (GDPR-specific holistic quality); and NLP metrics against LLM judge scores (agreement between the two automated assessment systems). Analyses are restricted to participants for whom data from all three sources (survey, NLP pipeline, judge evaluation) are available; the exact triangulated N is reported in the *Data Availability and Sample Sizes* subsection.

Statistical Analysis

A significance threshold of $\alpha = 0.05$ is applied to all tests. All within-subjects comparisons across the three modes follow a two-step procedure: a Friedman test as the omnibus test, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction for post-hoc analysis. Where the unbalanced full sample makes within-subjects matching impossible (e.g. when only one or two documents are available per participant), Mann–Whitney U tests are reported as a between-groups sensitivity check on the corresponding mode pair. Effect sizes are reported as ε^2 for the Friedman / Kruskal–Wallis omnibus tests, Kendall's W as the coefficient of concordance for Friedman, and rank-biserial correlation r for pairwise Wilcoxon signed-rank tests.

For RQ1 (User Experience and Preference), the Friedman test is applied to SUS scores, the R-TLX composite, and the confidence question composites across modes; post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction identify which mode pairs differ significantly; and mode preference distributions are reported as frequencies and rankings. For RQ2 (Document Quality and Confidence Calibration), the Friedman test is applied to NLP metric scores and LLM-as-a-judge scores across modes, with Bonferroni-corrected pairwise Wilcoxon signed-rank tests as post-hoc analysis where the omnibus is significant. Spearman rank correlation (ρ) is used for the confidence–quality gap analysis (non-parametric; appropriate for ordinal-to-continuous relationships).

Definitions of Statistical Tests and Effect Sizes

The following definitions apply to every measurement reported in the Evaluation chapter. They are listed once here rather than re-introduced for every instrument.

Shapiro–Wilk normality test. Tests the null hypothesis that a sample is drawn from a normal distribution. The statistic is

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \tag{3.10}$$

where $x_{(i)}$ are the order statistics and a_i are tabulated constants. W close to 1 indicates normality; small p-values reject normality. In this study, normality is rejected for nearly every survey measure, motivating the non-parametric tests below.

Friedman test. Non-parametric repeated-measures omnibus test for $k \geq 3$ related samples. Each participant's k values are ranked within-subject; the test statistic on n participants across k conditions is

$$\chi_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1), \tag{3.11}$$

where R_j is the sum of ranks for condition j. Under H_0 (no difference among conditions), χ_F^2 is asymptotically χ^2 distributed with $k-1$ degrees of freedom. A significant Friedman statistic motivates pairwise post-hoc comparisons.

Wilcoxon signed-rank test (post-hoc). Non-parametric test for two related samples, applied as a pairwise post-hoc to a significant Friedman omnibus. Differences $d_i = x_i - y_i$ are ranked by absolute value; the statistic

$$W = \min(W_+, W_-), \quad W_+ = \sum_{d_i>0} \mathrm{rank}(|d_i|), \quad W_- = \sum_{d_i<0} \mathrm{rank}(|d_i|). \tag{3.12}$$

With three modes there are three pairwise comparisons; the Bonferroni-corrected significance threshold is therefore $\alpha' = 0.05/3 = 0.0167$.

Mann–Whitney U test (between-groups sensitivity check). Non-parametric two-sample test on independent groups, used here as a sensitivity check on the unbalanced full sample where matched-pair tests are not applicable. With n_1 and n_2 the two group sizes and R_1 the sum of ranks in group 1,

$$U = R_1 - \frac{n_1(n_1+1)}{2}. \tag{3.13}$$

Kruskal–Wallis H test. Non-parametric one-way ANOVA analogue for k independent groups, used in the Evaluation chapter as a between-groups sensitivity check on the unbalanced full document sample. With N total observations, n_j observations in group j, and R_j the sum of ranks in group j,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1). \tag{3.14}$$

Chi-square goodness-of-fit test. Used here on the distribution of rank-1 votes in the mode-preference data, against a uniform null. With O_i observed and E_i expected counts in k categories,

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}. \tag{3.15}$$

Effect sizes. Effect sizes complement p-values by quantifying the magnitude of the difference, independent of sample size.

- Kendall's W (coefficient of concordance) is the Friedman test's effect size on $[0,1]$: $W = \chi^2_F/(n(k-1))$. Conventional bands: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.
- Rank-biserial correlation r is the matched-pairs effect size for the Wilcoxon signed-rank test, given by $r = (W_+ - W_-)/(W_+ + W_-)$. Conventional bands: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.
- Cramér's V is the chi-square effect size on $[0,1]$: $V = \sqrt{\chi^2/(N(k-1))}$. Conventional bands match Kendall's W.
- ε^2 (epsilon squared) is the Kruskal–Wallis effect size on $[0,1]$: $\varepsilon^2 = (H - k + 1)/(N - k)$.
- Spearman's ρ is the rank correlation coefficient on $[-1,+1]$, used for the confidence–quality analyses; magnitudes follow the standard correlation conventions (small $|\rho| \approx 0.1$, medium ≈ 0.3, large ≥ 0.5).

Bonferroni correction. A simple family-wise error-rate adjustment that divides the per-test significance threshold by the number of comparisons in the family. With m comparisons and a desired family-wise $\alpha = 0.05$, each individual test is evaluated against $\alpha' = \alpha/m$. With the three pairwise mode comparisons used throughout this work, $\alpha' = 0.0167$.

4 Evaluation and Results

This chapter reports the empirical findings of the user study and the subsequent document-quality analysis. The presentation follows the analysis plan laid out in Chapter 3: results are organised first around the per-mode self-report instruments (System Usability Scale, NASA Raw Task Load Index, Perceived AI Support), then the post-experiment mode preference ranking together with the open-ended free-text comments, and then the document-level quality measures (reference-based NLP metrics computed at the document level and the chapter level, followed by the LLM-as-a-judge evaluation). The chapter closes with a short overall summary and a descriptive Demographics section that profiles the participant sample. For every quantitative measure the per-mode results are reported in the order Form, Ask, Chat, followed by the aggregated value across all three modes; relative direction (increase, decrease, or comparable level) is stated explicitly. All inferential test statistics referenced in the running text are reproduced in full in Appendix E.

The survey analyses use the matched within-subjects sample of $n = 69$ participants. The document-quality analyses use the full set of 113 retained documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$) for descriptive statistics, and the strict matched-triple subset of $n = 26$ participants for within-subjects significance testing.

4.1 Usability — System Usability Scale

The System Usability Scale captures how usable each interface felt to participants immediately after they finished interacting with it. It consists of ten Likert items on a five-point scale, alternating in polarity: odd-numbered items are phrased positively (higher response equals more positive evaluation), and even-numbered items are phrased negatively (lower response equals more positive evaluation). This alternating keying is why the per-item table that follows must be read item by item rather than averaged at a glance. The full item text is reproduced in Appendix B. The aggregated "All" column in the tables below pools the three per-mode samples into a single per-item descriptive across all 207 ratings.

The ten SUS items capture distinct facets of usability — frequency of intended use, perceived complexity, ease of use, and so on — and the per-mode means therefore have to be read item by item before the aggregated test is reported. Table 4.1 reports the mean response per item per mode and the aggregated mean across all three modes. The aggregated column is the unweighted mean of the three per-mode means.

Table 4.1: System Usability Scale, mean response per item by mode ($n = 69$ per mode). Items are reported on the original five-point Likert scale; the "All" column averages the three per-mode means. Odd items: higher is better; even items: lower is better.

#	Item	Form	Ask	Chat	All
1	I think that I would like to use this system frequently.	3.52	3.22	3.22	3.32
2	I found the system unnecessarily complex.	1.64	2.25	2.12	2.00
3	I thought the system was easy to use.	4.38	4.01	3.99	4.13
4	I think that I would need the support of a technical person to be able to use this system.	1.52	1.49	1.55	1.52
5	I found the various functions in this system were well integrated.	4.00	3.67	3.64	3.77
6	I thought there was too much inconsistency in this system.	2.16	2.04	2.01	2.07
7	I would imagine that most people would learn to use this system very quickly.	4.30	4.00	4.03	4.11
8	I found the system very cumbersome to use.	1.65	2.16	2.29	2.03
9	I felt very confident using the system.	3.83	3.84	3.90	3.86
10	I needed to learn a lot of things before I could get going with this system.	1.61	1.61	1.62	1.61

For *"I think that I would like to use this system frequently"* (item 1), Form yields the highest mean (3.52), followed by Ask and Chat at the same level (3.22 each); the aggregated mean is 3.32. Frequent-use intent therefore decreases from Form to the two LLM-driven modes by approximately 0.30 points.

The negatively phrased complement to item 1 reverses the role of the modes. For *"I found the system unnecessarily complex"* (item 2, lower is better), Form is rated lowest (1.64), Chat second (2.12), and Ask highest at 2.25; the aggregated mean is 2.00. Perceived complexity is approximately 0.5 points higher in the LLM-driven modes than in Form.

The same Form-first ordering reappears on ease of use. For *"I thought the system was easy to use"* (item 3), Form is rated highest (4.38), with Ask and Chat closely matched at 4.01 and 3.99; the aggregated mean is 4.13. Ease-of-use ratings decrease by roughly 0.4 points from Form to the LLM-driven modes.

Item 4 then breaks the Form-first pattern by collapsing across modes. For *"I think that I would need the support of a technical person to be able to use this system"* (item 4, lower is better), the three modes are essentially indistinguishable (Form 1.52, Ask 1.49, Chat 1.55), with an aggregated mean of 1.52. None of the three modes is perceived to require technical support.

The Form-first ordering returns on the integration item. For *"I found the various functions in this system were well integrated"* (item 5), Form scores highest (4.00), Ask and Chat lower (3.67 and 3.64); aggregated mean is 3.77. The integration rating drops by approximately 0.35 points from Form to the LLM-driven modes.

As with the technical-support item, the inconsistency item levels out across modes. For *"I thought there was too much inconsistency in this system"* (item 6, lower is better), the three modes are similar (Form 2.16, Ask 2.04, Chat 2.01), aggregated mean 2.07. Form is rated marginally higher on this negative item than the two LLM-driven modes.

Echoing the ease-of-use item, learnability ratings again favour Form. For *"I would imagine that most people would learn to use this system very quickly"* (item 7), Form scores 4.30, Ask 4.00, Chat 4.03; aggregated mean is 4.11. Perceived learnability decreases by approximately 0.30 points from Form to the LLM-driven modes.

The widest gap of the ten items appears next. For *"I found the system very cumbersome to use"* (item 8, lower is better), the largest spread is observed: Form 1.65, Ask 2.16, Chat 2.29; aggregated mean 2.03. Cumbersomeness is rated approximately 0.5–0.65 points higher in the LLM-driven modes than in Form.

The confidence item then inverts the cumbersomeness ordering, albeit on a much narrower scale. For *"I felt very confident using the system"* (item 9), all three modes are rated similarly and slightly higher in the LLM-driven modes (Form 3.83, Ask

3.84, Chat 3.90); aggregated mean 3.86. Within-system confidence is broadly comparable across modes, with a small upward trend from Form to Chat.

The final item joins items 4 and 6 in showing no meaningful between-mode contrast. For *"I needed to learn a lot of things before I could get going with this system"* (item 10, lower is better), the three modes are virtually identical (Form 1.61, Ask 1.61, Chat 1.62), aggregated mean 1.61. The amount of prior learning required is perceived as low and equal across all modes.

Table 4.2 reports the full descriptive distribution per item per mode (mean, standard deviation, median, Q1, Q3). The minimum and maximum are 1 and 5 for every item $\times$ mode combination and are omitted for compactness.

The per-item distributions are visualised in Figure 4.1.

A Friedman test on the per-mode aggregate yields $\chi^2(2) = 5.51$, $p = 0.064$, $W = 0.040$, which does not reach the $\alpha = 0.05$ threshold. Pairwise Wilcoxon signed-rank post-hoc tests with Bonferroni correction ($\alpha = 0.0167$) indicate a significant Form-vs-Ask difference ($W = 578.5$, $p = 0.013$, rank-biserial $r = +0.368$, medium effect), a non-significant Form-vs-Chat difference ($p = 0.018$, just above the Bonferroni threshold), and no Ask-vs-Chat difference ($p = 0.666$). Full test statistics are reproduced in Appendix E.

Taken together, the SUS data establish a Form-first ordering on the items most directly tied to perceived ease of use, complexity, and cumbersomeness, while items relating to within-system confidence, required prior learning, and need for technical support flatten across modes. Whether this same ordering recurs on perceived workload is examined next in §4.2.

4.2 Cognitive Load — NASA Raw Task Load Index

Building on the SUS results reported in §4.1, this section turns to perceived workload. Where SUS measured how usable the interfaces felt, the Raw Task Load Index decomposes the participant's effort along six interpretable axes — Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration — each on a five-point Likert scale. Performance is reported in the positive-coded direction (higher equals worse perceived performance) for internal consistency with

Table 4.2: System Usability Scale, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `sus_per_item_summary.csv`.

#	Mode	Mean	SD	Median	Q1	Q3
1	All	3.32	1.14	3	3	4
	Form	3.52	1.09	4	3	4
	Ask	3.22	1.08	3	3	4
	Chat	3.22	1.24	3	2	4
2	All	2.00	1.12	2	1	3
	Form	1.64	0.92	1	1	2
	Ask	2.25	1.22	2	1	3
	Chat	2.12	1.12	2	1	3
3	All	4.13	0.97	4	4	5
	Form	4.38	0.91	5	4	5
	Ask	4.01	0.81	4	4	5
	Chat	3.99	1.13	4	3	5
4	All	1.52	0.89	1	1	2
	Form	1.52	0.85	1	1	2
	Ask	1.49	0.80	1	1	2
	Chat	1.55	1.01	1	1	2
5	All	3.77	1.05	4	3	5
	Form	4.00	0.92	4	4	5
	Ask	3.67	1.02	4	3	4
	Chat	3.64	1.16	4	3	5
6	All	2.07	1.00	2	1	3
	Form	2.16	0.96	2	1	3
	Ask	2.04	1.06	2	1	3
	Chat	2.01	0.98	2	1	3
7	All	4.11	0.98	4	4	5
	Form	4.30	0.96	5	4	5
	Ask	4.00	0.95	4	4	5
	Chat	4.03	1.01	4	4	5
8	All	2.03	1.13	2	1	3
	Form	1.65	0.94	1	1	2
	Ask	2.16	1.12	2	1	3
	Chat	2.29	1.23	2	1	3
9	All	3.86	1.04	4	3	5
	Form	3.83	1.04	4	3	5
	Ask	3.84	0.93	4	3	4
	Chat	3.90	1.15	4	4	5
10	All	1.61	0.90	1	1	2
	Form	1.61	0.86	1	1	2
	Ask	1.61	0.81	1	1	2
	Chat	1.62	1.03	1	1	2

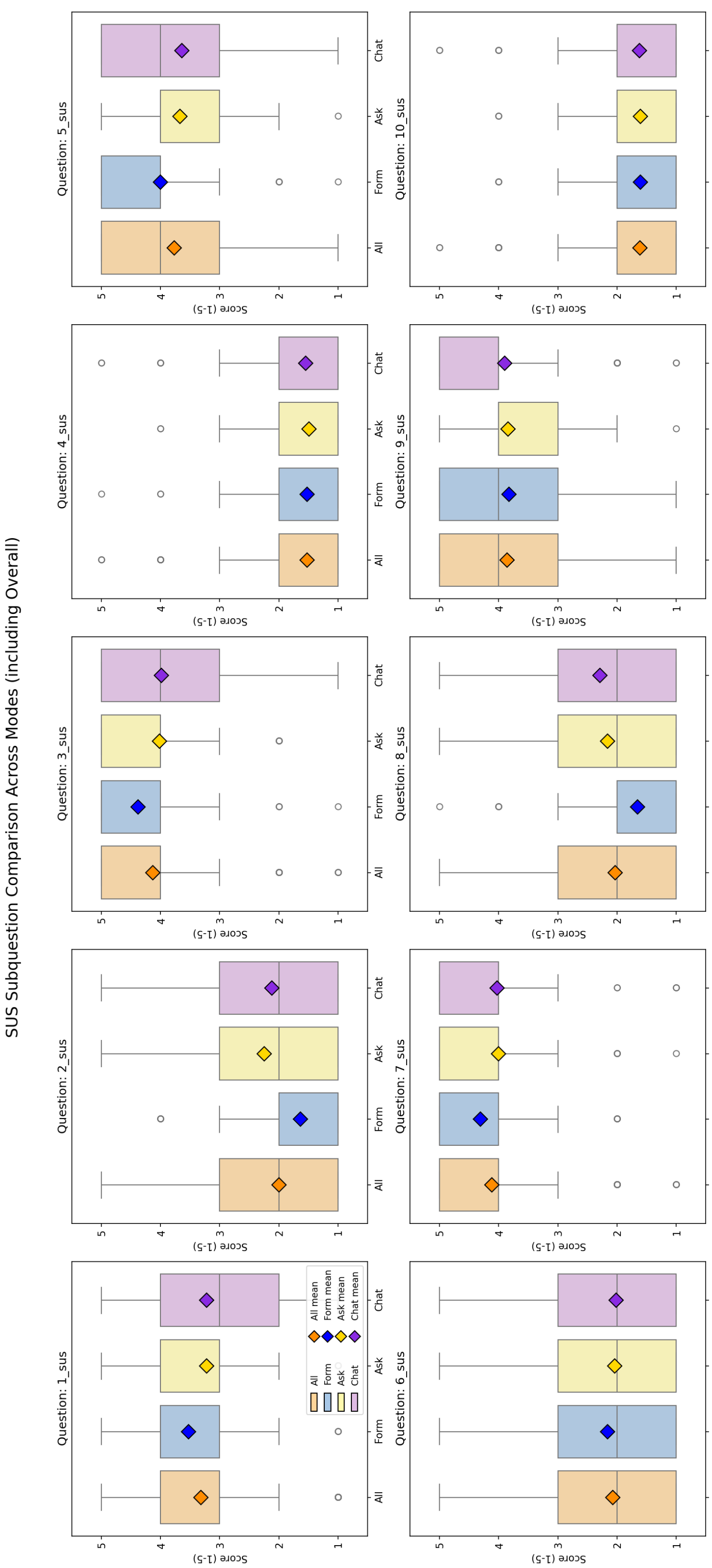

Figure 4.1: Box plots of the ten SUS items by mode ($n = 69$ per mode). Boxes show the interquartile range; the central line marks the median, and the diamond marks the mean.

the other subscales. The overall R-TLX in this study is the unweighted mean of the six subscales, rescaled to a 1–21 range. The expectation from the methodology framing is that subscales tied to interface mechanics (Mental Demand, Effort) will track the SUS ordering, while content-driven subscales (Temporal Demand, Frustration) may diverge where the LLM modes introduce additional variability. The full subscale descriptions are reproduced in Appendix B.

The six R-TLX subscales are reported individually before the aggregated test, so that the contribution of each axis to the overall workload score is visible. Table 4.3 reports the per-subscale mean and standard deviation per mode and the aggregated mean across modes. Figure 4.2 shows the corresponding distributions.

Table 4.3: NASA R-TLX subscale means $\pm$ standard deviation by mode ($n = 69$ per mode), 1–21 scale. The aggregated "All" column is the unweighted mean of the three per-mode means. All subscales are coded so that lower equals less workload.

Subscale	Form	Ask	Chat	All
Mental Demand	5.77 ± 4.23	6.80 ± 4.60	6.36 ± 5.27	6.31
Physical Demand	2.77 ± 2.69	3.54 ± 3.89	3.26 ± 3.43	3.19
Temporal Demand	4.16 ± 3.79	4.97 ± 4.12	5.54 ± 4.74	4.89
Effort	5.68 ± 4.53	6.39 ± 4.37	6.32 ± 4.74	6.13
Frustration	5.88 ± 4.80	7.58 ± 5.30	7.13 ± 5.41	6.86
Performance (pos.)	8.71 ± 5.44	8.68 ± 5.60	9.39 ± 6.32	8.93

Table 4.4 reports the full descriptive distribution per subscale per mode (mean, standard deviation, median, Q1, Q3, min, max). All subscales use the 1–21 scale where lower equals less workload (Performance is positive-coded for internal consistency).

The Form-first ordering observed throughout the SUS items (§4.1) recurs on the first workload subscale. For *Mental Demand*, Form scores lowest (mean 5.77), Chat next (6.36), and Ask highest (6.80); the aggregated mean is 6.31. Mental Demand increases by approximately 0.6–1.0 points when moving from Form to the LLM-driven modes.

By the same token, Physical Demand is lowest in Form, although the range is narrow. For *Physical Demand*, Form scores lowest (2.77), Chat second (3.26), Ask highest (3.54); aggregated 3.19. The absolute spread is small (≈ 0.8 points). Physical Demand is the lowest-rated subscale in all three modes.

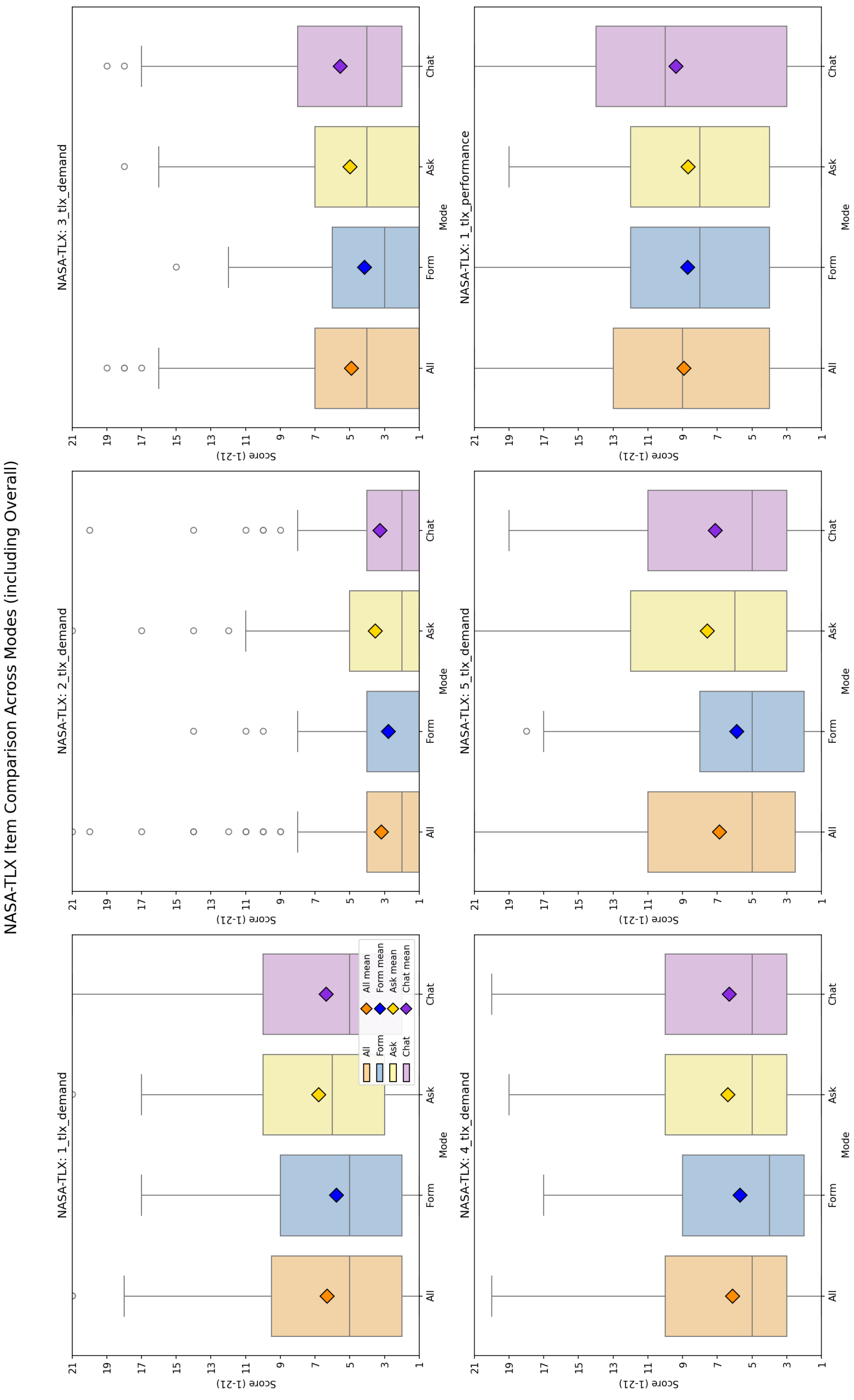

Figure 4.2: Box plots of the six R-TLX subscales by mode ($n = 69$ per mode).

Table 4.4: NASA R-TLX, full per-subscale descriptive distribution by mode ($n = 69$ per mode), 1–21 scale. Source: `tlx_per_item_summary.csv`.

Subscale	Mode	Mean	SD	Median	Q1	Q3	Min	Max
Mental Demand	All	6.31	4.75	5.0	2.0	9.5	1	21
	Form	5.77	4.26	5.0	2.0	9.0	1	17
	Ask	6.80	4.63	6.0	3.0	10.0	1	21
	Chat	6.36	5.31	5.0	2.0	10.0	1	21
Physical Demand	All	3.19	3.40	2.0	1.0	4.0	1	21
	Form	2.77	2.71	1.0	1.0	4.0	1	14
	Ask	3.54	3.92	2.0	1.0	5.0	1	21
	Chat	3.26	3.46	2.0	1.0	4.0	1	20
Temporal Demand	All	4.89	4.28	4.0	1.0	7.0	1	19
	Form	4.16	3.82	3.0	1.0	6.0	1	15
	Ask	4.97	4.15	4.0	1.0	7.0	1	18
	Chat	5.54	4.78	4.0	2.0	8.0	1	19
Effort	All	6.13	4.57	5.0	3.0	10.0	1	20
	Form	5.68	4.56	4.0	2.0	9.0	1	17
	Ask	6.39	4.41	5.0	3.0	10.0	1	19
	Chat	6.32	4.77	5.0	3.0	10.0	1	20
Frustration	All	6.86	5.24	5.0	2.5	11.0	1	21
	Form	5.88	4.84	5.0	2.0	8.0	1	18
	Ask	7.58	5.34	6.0	3.0	12.0	1	21
	Chat	7.13	5.44	5.0	3.0	11.0	1	19
Performance (pos.)	All	8.93	5.82	9.0	4.0	13.0	1	21
	Form	8.71	5.48	8.0	4.0	12.0	1	21
	Ask	8.68	5.64	8.0	4.0	12.0	1	19
	Chat	9.39	6.36	10.0	3.0	14.0	1	21

Temporal Demand sharpens the gap, departing from the Form-$<$-Chat-$<$-Ask shape seen above and turning monotonic. For *Temporal Demand*, the values increase monotonically from Form (4.16) to Ask (4.97) to Chat (5.54); aggregated 4.89. Temporal Demand is approximately 1.4 points higher in Chat than in Form, the largest monotonic increase observed across the six subscales.

Effort returns to the Form-versus-LLM split. For *Effort*, Form scores lowest (5.68), with Ask (6.39) and Chat (6.32) at a closely comparable level; aggregated 6.13. Effort increases by approximately 0.65 points from Form to the LLM-driven modes.

Frustration produces the widest absolute gap of the six subscales. For *Frustration*, Form scores lowest (5.88), Chat second (7.13), and Ask highest (7.58); aggregated 6.86. Frustration shows the largest absolute gap between Form and the LLM-driven modes ($+1.25$ to $+1.70$ points).

In contrast to the other five subscales, the self-rated Performance axis flattens almost completely. For *Performance* (positive-coded, higher equals worse perceived performance), Form and Ask are essentially identical (8.71 and 8.68); Chat is slightly higher at 9.39; aggregated 8.93. Perceived self-performance is comparable across modes, with a marginal degradation in Chat.

An aggregated Friedman test across the six subscales is non-significant ($\chi^2(2) = 5.24$, $p = 0.073$, $W = 0.038$). Among the six subscales, two reach significance: Temporal Demand ($\chi^2(2) = 12.77$, $p = 0.002$, $W = 0.093$) and Frustration ($\chi^2(2) = 8.93$, $p = 0.012$, $W = 0.065$). Pairwise post-hoc Wilcoxon signed-rank tests with Bonferroni correction identify Form-vs-Ask and Form-vs-Chat as significant on these two subscales, with medium-to-large rank-biserial effect sizes ($|r| = 0.36$ to 0.57); Ask versus Chat is non-significant on every measure. Full test statistics are reproduced in Appendix E.

The workload data therefore reinforce the Form-first ordering of the SUS, with the qualification that the gap is sharpest on Temporal Demand and Frustration and effectively absent on Performance. Whether participants nevertheless ascribe more value to the LLM-driven modes when asked specifically about AI support is the question taken up next in §4.3.

4.3 Perceived AI Support — AI Questionnaire

The SUS (§4.1) and R-TLX (§4.2) characterise the interface as it was experienced; they say nothing directly about whether participants felt the AI itself had been useful. The Perceived AI Support questionnaire targets exactly this gap. After each interaction mode, in addition to the SUS and R-TLX, participants completed a six-item Perceived AI Support questionnaire on a five-point Likert scale (1 = strong disagreement, 5 = strong agreement). The six items capture distinct facets of how participants perceived the AI's contribution to the document they had just created: helpfulness (item 1), efficiency (item 2), transparency (item 3), trust (item 4), legal-basis confidence (item 5), and re-use intent (item 6). Item 5 is the primary self-report confidence measure referenced in Chapter 3. The aggregated "All" column in the tables below pools the three per-mode samples into a single per-item descriptive across all 207 ratings. Given the consistent Form-first ordering on usability and workload, the question of interest here is whether perceived AI value follows the same pattern or breaks from it.

The six items are reported in their administered order, beginning with broad helpfulness and tightening towards specific facets such as transparency, trust, and re-use intent. Table 4.5 reports the mean response per item per mode and the aggregated mean across the three modes; the full descriptive distribution per item is reported in Table 4.6. Figure 4.3 shows the corresponding box-plot distributions.

Table 4.5: Perceived AI Support, mean response per item by mode ($n = 69$ per mode), five-point Likert scale (higher is better). The aggregated "All" column is the unweighted mean of the three per-mode means.

#	Item	Form	Ask	Chat	All
1	The AI was helpful in creating this document.	4.23	4.32	4.13	4.23
2	The AI helped me to complete the document faster.	4.43	4.25	3.99	4.22
3	I understood why the AI suggested specific GDPR fields.	3.51	4.16	3.75	3.81
4	I trust that the AI made correct suggestions.	3.41	3.80	3.68	3.63
5	I am confident that the AI identified the correct legal basis.	3.58	3.74	3.81	3.71
6	I would use this mode again.	3.99	3.55	3.42	3.65

Independently of the usability ordering, perceived helpfulness favours Ask. For *"The AI was helpful in creating this document"* (item 1), Form scores 4.23, Ask 4.32, Chat

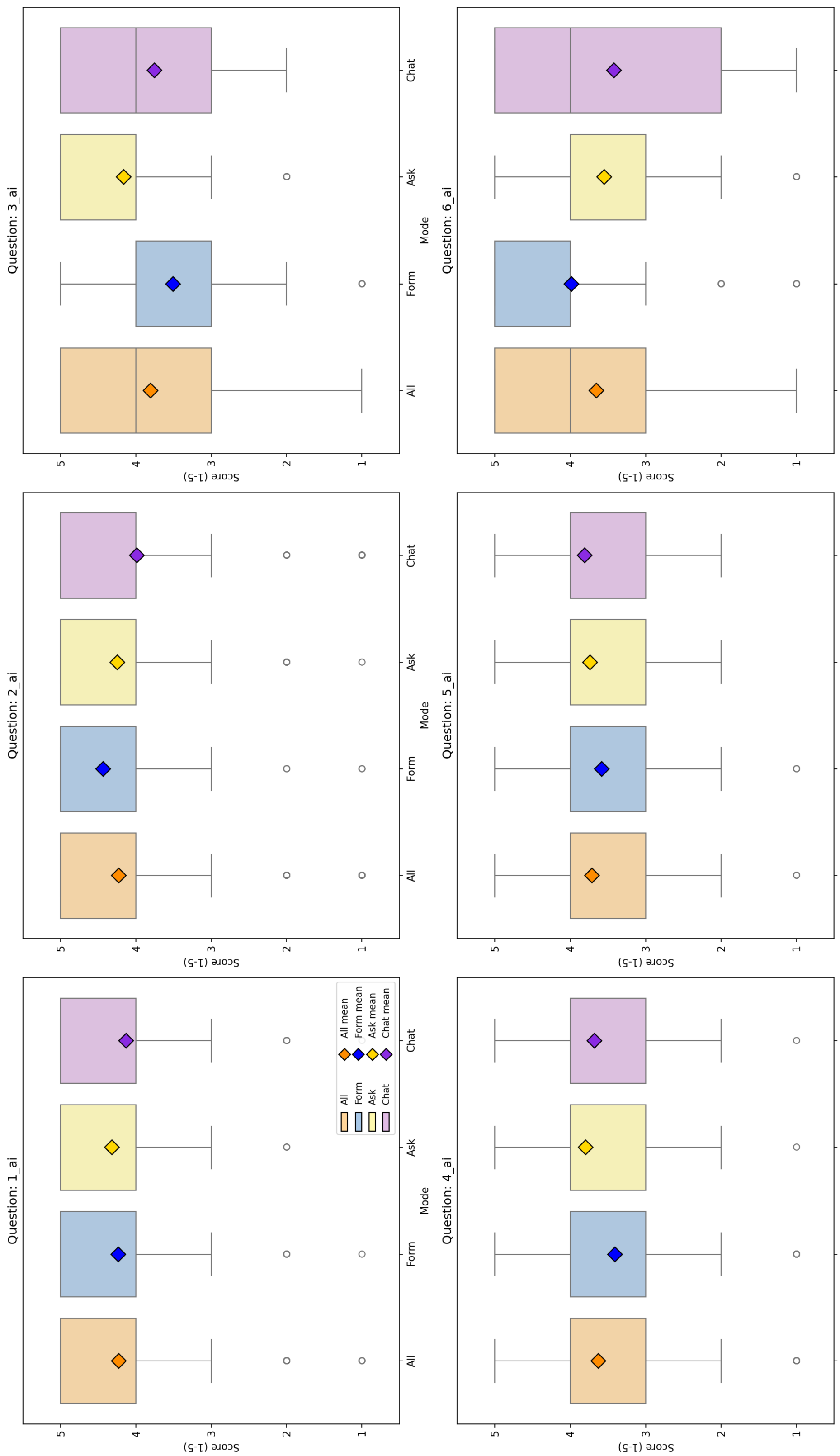

Figure 4.3: Box plots of the six Perceived AI Support items by mode ($n = 69$ per mode). Boxes show the interquartile range; the central line marks the median, and the diamond marks the mean.

4.13; aggregated 4.23. Helpfulness is rated highest in Ask and lowest in Chat, with a small absolute spread of roughly 0.20 points.

The picture changes when efficiency is asked about explicitly. For *"The AI helped me to complete the document faster"* (item 2), Form is rated highest (4.43), Ask second (4.25), Chat lowest (3.99); aggregated 4.22. Perceived speed-up decreases monotonically with increasing LLM involvement, dropping by approximately 0.45 points from Form to Chat.

Transparency is the first item on which the LLM-driven modes overtake Form, and the gap is the widest of the six items. For *"I understood why the AI suggested specific GDPR fields"* (item 3), Form scores lowest (3.51), Chat second (3.75), Ask highest (4.16); aggregated 3.81. Transparency is rated approximately 0.65 points higher in Ask than in Form. Ask is the only mode that explicitly explains each suggestion.

Echoing the transparency ordering, trust also rises with LLM involvement. For *"I trust that the AI made correct suggestions"* (item 4), Form scores lowest (3.41), Chat second (3.68), Ask highest (3.80); aggregated 3.63. Trust increases by approximately 0.27–0.39 points from Form to the LLM-driven modes.

The primary self-report confidence measure follows the same direction. For *"I am confident that the AI identified the correct legal basis"* (item 5, primary confidence measure), the three modes show a monotonic increase: Form 3.58, Ask 3.74, Chat 3.81; aggregated 3.71. Confidence in legal correctness rises by approximately 0.23 points from Form to Chat.

This pattern is reversed for re-use intent, returning to the Form-first ordering of the SUS (§4.1). For *"I would use this mode again"* (item 6), Form scores highest (3.99), Ask second (3.55), Chat lowest (3.42); aggregated 3.65. Re-use intent decreases by approximately 0.57 points from Form to Chat, the largest negative gap of the six items.

Beyond the per-mode means, the full descriptive distribution shows how concentrated or spread participants' responses were on each item. Table 4.6 reports the mean, standard deviation, median, Q1, and Q3 per item per mode. The minimum is 1 and the maximum is 5 for every cell (both endpoints of the Likert scale are realised in every item $\times$ mode combination), so they are omitted from the table for compactness.

Table 4.6: Perceived AI Support, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `ai_support_per_item_summary.csv`.

#	Mode	Mean	SD	Median	Q1	Q3
1	All	4.23	0.91	4	4	5
	Form	4.23	0.97	5	4	5
	Ask	4.32	0.76	4	4	5
	Chat	4.13	1.00	4	4	5
2	All	4.22	1.13	5	4	5
	Form	4.43	0.96	5	4	5
	Ask	4.25	1.03	5	4	5
	Chat	3.99	1.32	4	4	5
3	All	3.81	1.06	4	3	5
	Form	3.51	1.17	4	3	4
	Ask	4.16	0.92	4	4	5
	Chat	3.75	0.98	4	3	5
4	All	3.63	0.97	4	3	4
	Form	3.41	1.09	4	3	4
	Ask	3.80	0.87	4	3	4
	Chat	3.68	0.92	4	3	4
5	All	3.71	0.88	4	3	4
	Form	3.58	0.90	4	3	4
	Ask	3.74	0.87	4	3	4
	Chat	3.81	0.88	4	3	4
6	All	3.65	1.25	4	3	5
	Form	3.99	1.14	4	4	5
	Ask	3.55	1.16	4	3	4
	Chat	3.42	1.39	4	2	5

An aggregated Friedman test on the per-mode aggregate is non-significant ($\chi^2(2) = 1.05$, $p = 0.592$, $W = 0.008$), indicating that perceived AI support does not differ meaningfully across modes when the six items are considered together. The single legal-basis confidence item 5_{ai} is also non-significant under Friedman ($\chi^2(2) = 3.76$, $p = 0.152$). The re-use item 6_{ai} does reach significance ($\chi^2(2) = 11.12$, $p = 0.004$, $W = 0.081$), with a large rank-biserial effect for Form versus Chat ($r = +0.543$). Full statistics are reproduced in Appendix E.

Perceived AI support thus splits the six items into two camps: items that frame the AI as an explainer (transparency, trust, legal-basis confidence) favour the LLM-driven modes, while items that frame it as a productivity aid (efficiency, re-use intent) favour Form. The behavioural counterpart of these per-item ratings — which mode participants actually choose when asked to rank — is examined next in §4.4.

4.4 Mode Preference and Free-Text Feedback

The per-instrument data so far have shown a consistent Form-first ordering on usability (§4.1) and workload (§4.2), with a partial inversion on perceived transparency, trust, and confidence (§4.3). This section asks how the three patterns combine when participants are forced to commit to a single overall preference. After all three interaction modes had been completed, participants provided two final inputs: an overall ranking of the three modes from most to least preferred, and an open-ended summary comment on the study experience. In addition, after each individual mode, an optional per-mode comment was collected. This section reports the ranking distribution, the inferential test on the rank-1 votes, and a thematic summary of all 88 free-text comments contributed by 46 unique participants.

4.4.1 Mode Preference Ranking

The ranking offers the cleanest test of which mode participants actually preferred once they had used all three. Table 4.7 summarises the post-experiment ranking. Participants assigned each of the three modes a rank from 1 (most preferred) to 3 (least preferred); “Rank-1 votes” counts the participants who placed the mode first.

Table 4.7: Post-experiment mode preference ranking ($n = 69$). "Rank-1 votes" counts the number of participants who placed the mode first; "Share" is the corresponding percentage; "Mean rank" averages the assigned 1–3 rank across all 69 participants.

Mode	Rank-1 votes	Share	Mean rank
Form	37	53.6 %	1.78
Chat	20	29.0 %	2.07
Ask	12	17.4 %	2.14

Form receives the plurality of first-place votes ($53.6\,\%$), followed by Chat ($29.0\,\%$) and Ask ($17.4\,\%$). The mean ranks follow the same ordering, mirroring the Form-first pattern of the SUS and R-TLX. A chi-square goodness-of-fit test against a uniform null rejects the null hypothesis ($\chi^2(2, N = 69) = 14.17, p < 0.001$, Cramér's $V = 0.320$, medium effect). The complementary Friedman test on the full rank distribution is non-significant ($\chi^2(2) = 5.07$, $p = 0.079$), reflecting that the strong Form plurality at rank 1 is partially balanced by variance at ranks 2 and 3. Full statistics are reproduced in Appendix E.

The complete distribution of all three rank positions across the three modes is visualised in Figure 4.4.

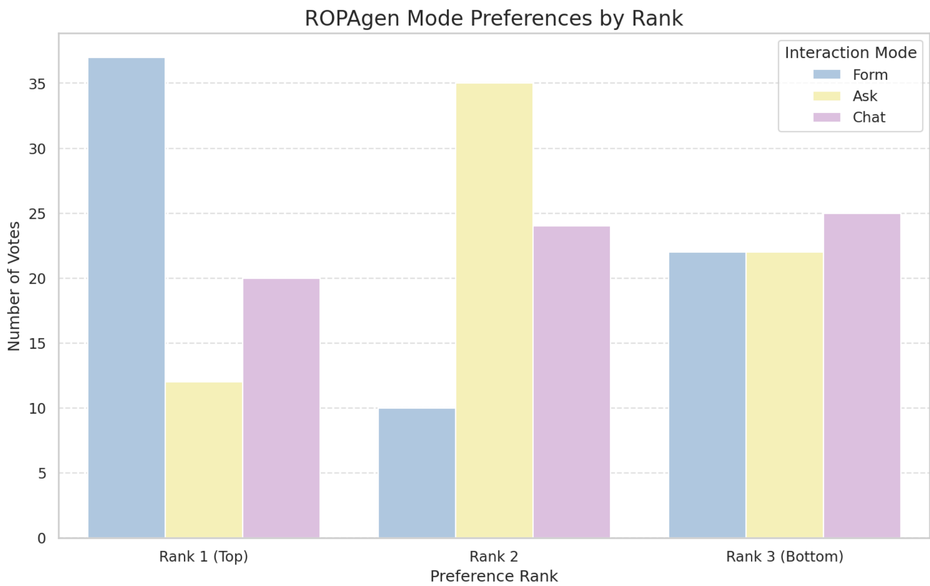

Figure 4.4: Distribution of all three rank positions per mode ($n = 69$).

4.4.2 Free-Text Comments

The free-text comments contextualise the ranking by surfacing the reasons participants gave for their preferences. Each mode collected an optional per-mode open-ended comment, and after all three modes participants wrote an overall summary comment. A total of 88 comments were recorded across 46 unique participants. Comments are written in German with a small minority in English; they are summarised thematically below. The presentation preserves the order Form, Ask, Chat, and concludes with the across-modes overall comments.

Form. Comments on Form cluster around speed, clarity, and control. Recurring observations describe the mode as the fastest of the three, the easiest to obtain an overview of, and the most "integrated" into the document creation workflow because the checkbox interface makes the selection space visible at a glance. Several participants emphasise that Form "takes the most work off you" (*"nimmt einem die meiste Arbeit ab"*) by pre-populating the option set, while a smaller subset notes the opposite concern: that the absence of an interactive AI commentary makes it unclear on what basis the model produces its suggestions, and that participants would not feel safe relying on Form output without an additional verification step. Several comments describe specific UX issues: long text fields rendered as a single narrow line on mobile, the inability to provide the AI with additional context for its suggestions, and the impression that the AI "guesses" without scenario-specific input.

Ask. Comments on Ask are the most polarised of the three modes. A consistent theme is that Ask is the most cumbersome mode in terms of workflow: the requirement to manually copy the AI's suggested text into the form field is repeatedly described as redundant work, particularly because participants reported that the model often produced suggestions in a free-text style that did not match the structured form fields. Several participants observed that the lack of a persistent context across fields forced them to re-enter the same information repeatedly. Conversely, explanatory text generated by Ask is consistently described as the most informative of the three modes, supporting the high transparency rating visible in item 3 of the

AI Support questionnaire. A frequent suggestion is to combine Ask's explanations with Form's structured field layout.

Chat. Comments on Chat are dominated by two themes: comfort and opacity. On the comfort side, several participants describe Chat as the most intuitive of the three modes, especially because the AI takes over the work of mapping the conversation onto the structured ROPA fields, leaving the participant only with the task of describing the processing scenario. On the opacity side, an equally large group of participants reports uncertainty about what is actually being written into the document — whether the entire chat history is used for generation, whether each individual message must be confirmed, and what the document looks like during the conversation. Specific defects mentioned include the AI occasionally switching from German to English mid-response, occasional emission of raw `DATA_UPDATE` tokens, and a slow "streaming" rendering style that participants experienced as artificially throttled.

Across modes (overall comments). The overall comments recapitulate the per-mode pattern. A clear majority describes Form as the fastest and most ergonomic mode, particularly for participants who feel they "do not know enough" to interrogate the model meaningfully. Chat is typically placed second and motivated by the impression that "the AI takes care of it," which several participants explicitly note increases their *subjective* sense of safety even though they cannot verify the output. Ask is most often placed third, with the manual copy-paste step cited as the principal reason. A recurring constructive suggestion across participants is a hybrid mode that combines the visible option set of Form with the conversational, context-aware behaviour of Chat or Ask. Several participants explicitly add a caveat that they are novices in the legal domain and that their preference ordering might shift if they had prior GDPR knowledge.

The ranking and comment data therefore close the user-experience half of the chapter on the same Form-first ordering established by the SUS and R-TLX, with Chat slotting in second on raw votes and Ask trailing largely because of its copy-paste workflow. Whether the documents produced under each mode mirror this ordering — or invert it — is the question taken up next, beginning with the reference-based NLP metrics in §4.5.

4.5 Document Quality — Reference-Based NLP Metrics

The user-experience half of the chapter (§4.1–§4.4) showed a stable Form-first ordering on usability, workload, and overall preference, with Chat second and Ask third. This section opens the document- quality half by asking whether the same ordering holds when the output text itself is compared against a reference document. Document-quality results are reported on the full retained set of 113 documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$) for descriptive statistics, and on the strict matched-triple subset of $n = 26$ for within-subjects significance tests. The metric suite mirrors the one applied by von Schwerin et al. [14] in their evaluation of LLM-generated ROPA data categories. All metrics are computed against the single expert-authored Form-mode reference document described in Chapter 3. For each metric, the per-mode results are reported in the order Form, Ask, Chat, followed by the aggregated value across all three modes. The descriptive tables in this section report — per metric and per mode — the full empirical distribution (n, min, Q1, median, Q3, max, mean, SD) computed from the raw per-document scores in the pipeline outputs `document_results.csv` and `chapter_results.csv`.

4.5.1 Document-Level Results

This subsection reports each metric over the full participant document, treating the ROPA as a single block of text.

Lexical Metrics

Lexical metrics — BLEU, ROUGE, and METEOR — count surface-form token overlap with the reference document; they reward word-level fidelity but are blind to paraphrase, and are therefore reported first because they make the strictest demands on surface form. An LLM-generated document that covers the same content with different wording will score lower here than under the embedding-based metrics that follow. Table 4.8 reports the full distribution of the five lexical metrics

(BLEU, ROUGE-1/2/L, METEOR) per mode, plus the aggregated row across all 113 documents. Figure 4.5 shows the corresponding box-plot distributions.

BLEU. BLEU *(Bilingual Evaluation Understudy)* ((3.1)) counts exact n-gram overlap with the reference and is therefore harsh on paraphrase: a document that says the same thing in different words scores low. It is precision-oriented (how much of what the candidate said also appears in the reference) with a brevity penalty, and the values reported here use $N = 4$ (matching unigrams through 4-grams).

Unlike the user-experience instruments, BLEU does not reproduce the Form-first ordering. For *BLEU*, Form yields a mean of 0.129, Ask 0.127, Chat 0.143; the aggregated mean is 0.133. Chat scores marginally higher than Form and Ask, which are essentially indistinguishable. The aggregated Friedman test on the matched subset is significant ($\chi^2(2) = 6.23$, $p = 0.044$); the significant pairwise difference is Chat $>$ Ask ($r = -0.578$, large effect).

ROUGE-1, ROUGE-2, ROUGE-L. ROUGE *(Recall-Oriented Understudy for Gisting Evaluation)* ((3.2)–(3.4)) measures how much of the reference's wording the candidate recovers, in three flavours: unigrams (single words), bigrams (adjacent word pairs), and the longest common subsequence. Each is reported as F1 — the harmonic mean of precision (how much of what the candidate said belongs) and recall (how much of what should have been said is there) — so it is more permissive than BLEU in word order but still surface-form.

As with BLEU, ROUGE-1 places Ask last rather than producing the Form-first ordering of the user-experience data. For *ROUGE-1*, Form yields 0.488, Ask 0.465, Chat 0.511; aggregated 0.488. ROUGE-1 increases from Ask to Form to Chat. The aggregated Friedman test is significant ($\chi^2(2) = 10.69$, $p = 0.005$); Form $>$ Ask and Chat $>$ Ask are significant pairwise differences with large effects, while Form versus Chat is not significant.

The picture changes at the bigram level, where Form takes the lead. For *ROUGE-2*, Form yields the highest mean (0.306), Chat second (0.291), Ask lowest (0.257); aggregated 0.285. ROUGE-2 follows the Form $>$ Chat $>$ Ask ordering. The aggregated Friedman test is significant ($\chi^2(2) = 13.46$, $p = 0.001$); Form $>$ Ask is significant ($r = +0.829$, large effect), as is Chat $>$ Ask.

Table 4.8: Document-level lexical metric distributions by mode (full sample $n = 113$). Source: `doc_nlp_summary.csv`.

Metric	Mode	n	Min	Q1	Median	Q3	Max	Mean	SD
BLEU	Form	37	0.064	0.088	0.124	0.158	0.230	0.129	0.048
	Ask	37	0.047	0.083	0.125	0.148	0.246	0.127	0.056
	Chat	39	0.067	0.122	0.134	0.163	0.235	0.143	0.039
	All	113	0.047	0.094	0.128	0.159	0.246	0.133	0.048
ROUGE-1	Form	37	0.313	0.421	0.493	0.569	0.644	0.488	0.098
	Ask	37	0.322	0.412	0.449	0.522	0.618	0.465	0.088
	Chat	39	0.330	0.468	0.517	0.561	0.645	0.511	0.071
	All	113	0.313	0.425	0.491	0.560	0.645	0.488	0.087
ROUGE-2	Form	37	0.178	0.241	0.301	0.361	0.455	0.306	0.076
	Ask	37	0.121	0.200	0.238	0.275	0.429	0.257	0.080
	Chat	39	0.131	0.253	0.283	0.344	0.447	0.291	0.069
	All	113	0.121	0.234	0.268	0.344	0.455	0.285	0.077
ROUGE-L	Form	37	0.238	0.320	0.391	0.468	0.566	0.388	0.094
	Ask	37	0.194	0.296	0.326	0.376	0.545	0.347	0.091
	Chat	39	0.230	0.348	0.387	0.444	0.544	0.389	0.072
	All	113	0.194	0.311	0.359	0.441	0.566	0.375	0.087
METEOR	Form	37	0.335	0.367	0.419	0.442	0.514	0.412	0.051
	Ask	37	0.234	0.327	0.366	0.409	0.488	0.365	0.068
	Chat	39	0.235	0.374	0.389	0.421	0.501	0.390	0.057
	All	113	0.234	0.359	0.388	0.431	0.514	0.389	0.062

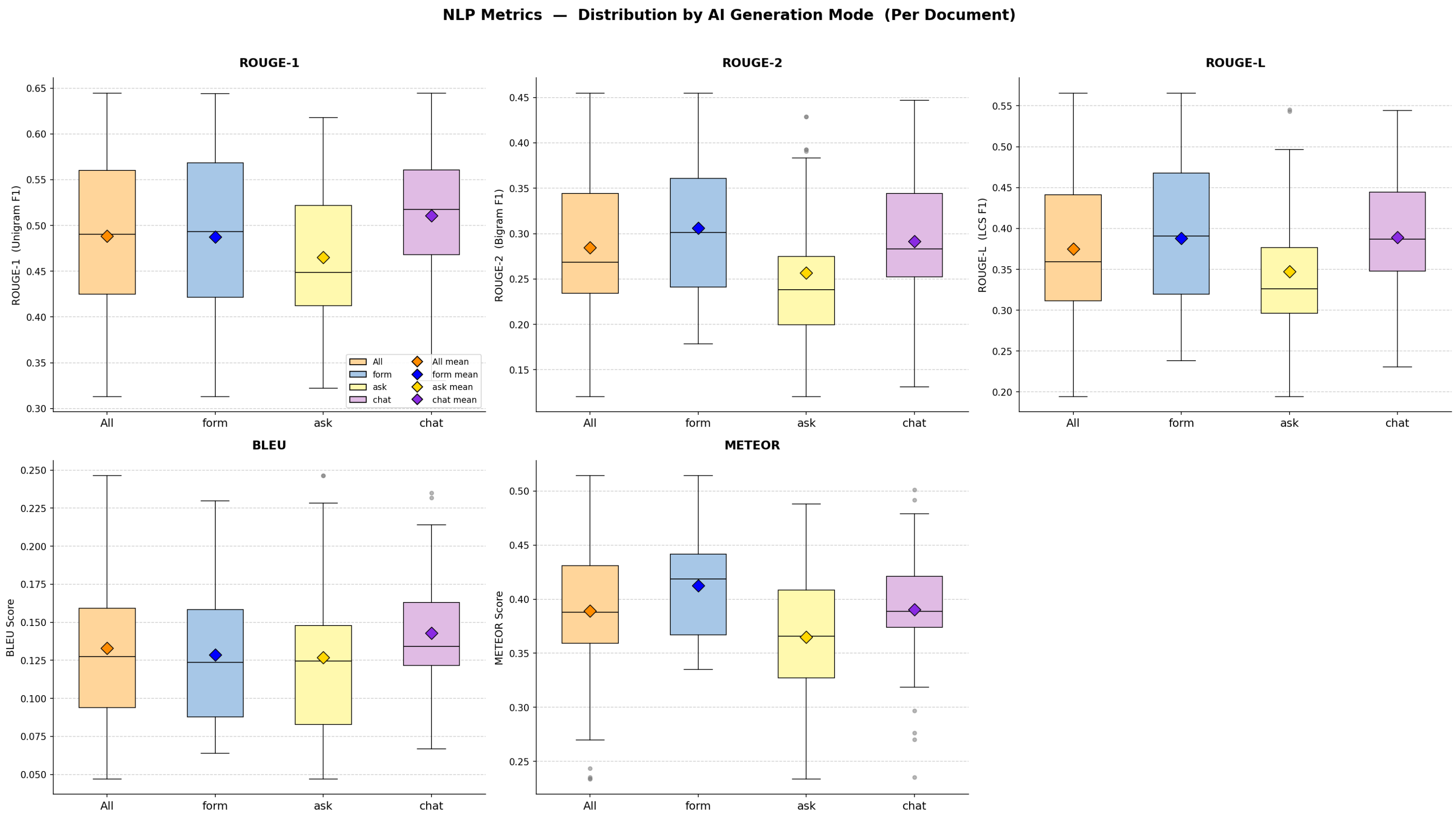

Figure 4.5: Document-level distributions of BLEU, ROUGE-1/2/L, and METEOR by mode (full sample $n = 113$). Boxes show the interquartile range; the diamond marks the mean.

ROUGE-L returns to the BLEU/ROUGE-1 shape with Form and Chat co-leading. For *ROUGE-L*, Form (0.388) and Chat (0.389) are essentially tied, with Ask trailing (0.347); aggregated 0.375. The aggregated Friedman test is significant ($\chi^2(2) = 11.31$, $p = 0.004$); Form $>$ Ask and Chat $>$ Ask are significant pairwise differences with large effects.

METEOR. METEOR *(Metric for Evaluation of Translation with Explicit ORdering)* ((3.5)) softens BLEU by allowing matches not just on identical tokens but also on stems and WordNet synonyms, so a document that paraphrases with related words is no longer punished as if it had said something unrelated. It is a fragmentation-penalised harmonic mean of unigram precision and recall, sitting between BLEU and ROUGE in strictness.

As with ROUGE-2, METEOR favours Form. For *METEOR*, Form yields the highest mean (0.412), Chat second (0.390), Ask lowest (0.365); aggregated 0.389. The aggregated Friedman test is significant ($\chi^2(2) = 9.92$, $p = 0.007$); Form $>$ Ask is significant ($r = +0.812$, large effect), and the remaining pairwise differences do not reach the Bonferroni threshold.

BERT-Based and Sentence-Embedding Metrics

BERT-based metrics measure semantic rather than lexical similarity, comparing contextual embeddings rather than literal tokens. Where the lexical metrics penalise paraphrase, the embedding-based metrics credit semantic overlap and therefore offer a complementary view of the same documents. Table 4.9 reports the same full distribution for the BERT-based metrics (BERTScore Precision / Recall / F1) and the document-level SBERT cosine similarity computed with ModernBERT embeddings. Figure 4.6 shows the corresponding box-plot distributions.

BERTScore Precision, Recall, F1. BERTScore ((3.6)–(3.8)) replaces n-gram matching with cosine similarities between contextual token embeddings, so paraphrase that shares meaning is credited even without surface-form overlap. *Precision* measures how much of what the candidate said belongs (penalising spurious

Table 4.9: Document-level BERT-based and SBERT distributions by mode (full sample $n = 113$). Source: `doc_bert_summary.csv`.

Metric	Mode	n	Min	Q1	Median	Q3	Max	Mean	SD
BERTScore Precision	Form	37	0.862	0.880	0.889	0.899	0.912	0.889	0.014
	Ask	37	0.861	0.872	0.889	0.899	0.918	0.886	0.016
	Chat	39	0.867	0.890	0.896	0.902	0.922	0.895	0.012
	All	113	0.861	0.880	0.891	0.900	0.922	0.890	0.014
BERTScore Recall	Form	37	0.914	0.919	0.923	0.926	0.936	0.923	0.005
	Ask	37	0.861	0.908	0.915	0.918	0.925	0.908	0.017
	Chat	39	0.874	0.908	0.917	0.922	0.933	0.913	0.013
	All	113	0.861	0.914	0.918	0.922	0.936	0.915	0.014
BERTScore F1	Form	37	0.887	0.900	0.906	0.912	0.920	0.905	0.009
	Ask	37	0.863	0.892	0.896	0.906	0.918	0.897	0.013
	Chat	39	0.872	0.899	0.906	0.911	0.922	0.904	0.010
	All	113	0.863	0.895	0.904	0.910	0.922	0.902	0.011
SBERT (ModernBERT)	Form	37	0.974	0.984	0.988	0.989	0.992	0.986	0.004
	Ask	37	0.971	0.983	0.987	0.990	0.993	0.986	0.005
	Chat	39	0.985	0.988	0.990	0.991	0.994	0.990	0.002
	All	113	0.971	0.985	0.988	0.990	0.994	0.987	0.004

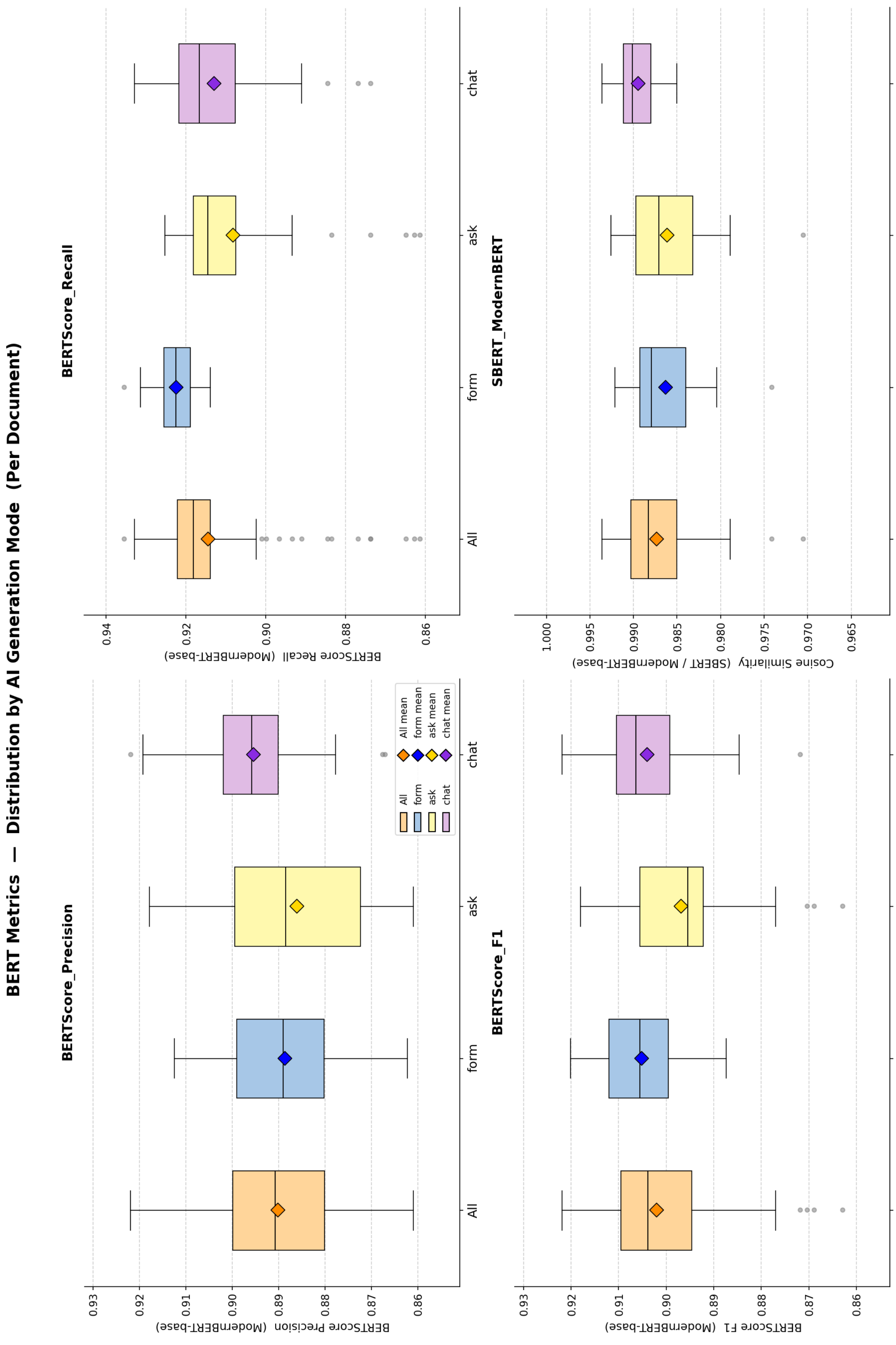

Figure 4.6: Document-level distributions of BERTScore Precision / Recall / F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$).

content), *recall* measures how much of what should have been said is there (penalising omission), and *F1* is their harmonic mean — both must be reasonable for F1 to be high, since the harmonic mean is dragged down by the smaller of the two.

BERTScore Precision puts Chat first, breaking with the lexical ordering above. For *BERTScore Precision*, Chat yields the highest mean (0.895), Form second (0.889), Ask lowest (0.886); aggregated 0.890. The aggregated Friedman test is significant ($\chi^2(2) = 18.54$, $p < 0.001$); Form $>$ Ask and Chat $>$ Ask are significant.

This pattern is reversed for Recall, which produces the largest effect of the entire metric pipeline. For *BERTScore Recall*, Form yields the highest mean (0.923), with Chat (0.913) and Ask (0.908) trailing; aggregated 0.915. The aggregated Friedman test is significant ($\chi^2(2) = 30.77$, $p < 0.001$, $W = 0.410$, the largest effect observed in the metric pipeline); Form $>$ Ask is significant with the maximum possible matched-pairs effect ($r = +1.000$), and Form $>$ Chat is also significant ($r = +0.687$, large effect). Ask versus Chat is non-significant.

Combining the two yields a near-tie at the top. For *BERTScore F1*, Form yields the highest mean (0.905), Chat second (0.904), Ask lowest (0.897); aggregated 0.902. The aggregated Friedman test is significant ($\chi^2(2) = 13.00$, $p = 0.002$); Form $>$ Ask and Chat $>$ Ask are significant with large effects. Form versus Chat is non-significant on F1, reflecting the trade-off between Form's higher Recall and Chat's higher Precision.

SBERT (ModernBERT) cosine similarity. Sentence-BERT cosine similarity ((3.9)) compresses each document into a single embedding vector and reports how aligned the two vectors are, on a scale where 1 is identical meaning and 0 is unrelated. It is the most lenient of the metrics reported here: because the comparison is at the document level rather than the token level, two documents that cover the same topics with different wording typically obtain values close to 1.

At the document-embedding level, Chat takes a clear lead. For *SBERT cosine similarity (ModernBERT)*, Chat yields the highest mean (0.990), with Form and Ask tied at 0.986; aggregated 0.987. The aggregated Friedman test is significant ($\chi^2(2) = 16.69$, $p < 0.001$); Chat $>$ Form and Chat $>$ Ask are both significant with large effects.

In summary, on the document level the three modes split into the patterns *Form* $\approx$ *Chat* $>$ *Ask* (BLEU, ROUGE-1, ROUGE-L, METEOR, BERTScore Precision, BERTScore F1), *Form* $>$ *Chat* $>$ *Ask* on n-gram overlap of size two and BERTScore Recall, and *Chat* $>$ *Form* $\approx$ *Ask* on SBERT. Ask underperforms both Form and Chat on every metric; Form and Chat trade leadership depending on whether the metric emphasises lexical recall (Form leads) or semantic-document-level fluency (Chat leads). The full within-subjects test results are reproduced in Appendix E.

4.5.2 Chapter-Level Results

This subsection reports the same metrics computed per ROPA chapter, so that scores are not dominated by long sections relative to short ones. Document-level scores collapse the entire ROPA into a single number, which is convenient for ranking modes but obscures where in the document the differences originate. The chapter-level decomposition addresses this by recomputing the same nine metrics at the level of the six Article 30(1) ROPA sections — *Verantwortliche Stelle und Kontaktdaten* (Controller and Contact Details), *Zwecke und Rechtsgrundlagen der Verarbeitung* (Purposes and Legal Bases of Processing), *Kategorien personenbezogener Daten und Datenquellen* (Categories of Personal Data and Sources), *Betroffene Personen und Empfänger* (Data Subjects and Recipients), *Aufbewahrungsfristen und Löschung* (Retention Periods and Deletion), and *Technische und organisatorische Maßnahmen* (Technical and Organisational Measures). Chapter-level results characterise where in the document each mode performs best. Tables 4.10–4.18 report mean (SD) per chapter per mode for each of the nine metrics.

Table 4.10: Chapter-level BLEU mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.179 (0.077)	0.200 (0.085)	0.166 (0.076)	0.181 (0.080)
Purposes & Legal Bases	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Personal Data Categories	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Data Subjects & Recipients	0.013 (0.010)	0.008 (0.014)	0.014 (0.015)	0.012 (0.014)
Retention & Deletion	0.038 (0.044)	0.023 (0.073)	0.068 (0.094)	0.044 (0.076)
Technical & Org. Measures	0.078 (0.068)	0.070 (0.080)	0.091 (0.047)	0.080 (0.066)

Table 4.11: Chapter-level ROUGE-1 mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.550 (0.097)	0.605 (0.129)	0.559 (0.109)	0.571 (0.114)
Purposes & Legal Bases	0.029 (0.014)	0.031 (0.024)	0.024 (0.012)	0.028 (0.017)
Personal Data Categories	0.056 (0.035)	0.082 (0.062)	0.071 (0.045)	0.070 (0.049)
Data Subjects & Recipients	0.090 (0.034)	0.093 (0.060)	0.123 (0.052)	0.103 (0.052)
Retention & Deletion	0.252 (0.092)	0.212 (0.132)	0.314 (0.143)	0.260 (0.131)
Technical & Org. Measures	0.313 (0.137)	0.330 (0.157)	0.350 (0.090)	0.331 (0.130)

Table 4.12: Chapter-level ROUGE-2 mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.363 (0.131)	0.430 (0.158)	0.365 (0.138)	0.386 (0.145)
Purposes & Legal Bases	0.014 (0.013)	0.012 (0.016)	0.008 (0.009)	0.011 (0.013)
Personal Data Categories	0.035 (0.030)	0.053 (0.046)	0.036 (0.029)	0.041 (0.036)
Data Subjects & Recipients	0.072 (0.030)	0.059 (0.056)	0.089 (0.049)	0.073 (0.048)
Retention & Deletion	0.134 (0.071)	0.087 (0.107)	0.168 (0.137)	0.131 (0.113)
Technical & Org. Measures	0.166 (0.120)	0.163 (0.141)	0.197 (0.089)	0.176 (0.118)

Table 4.13: Chapter-level ROUGE-L mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.518 (0.110)	0.576 (0.140)	0.511 (0.128)	0.534 (0.129)
Purposes & Legal Bases	0.028 (0.015)	0.030 (0.022)	0.024 (0.012)	0.027 (0.017)
Personal Data Categories	0.056 (0.035)	0.082 (0.062)	0.071 (0.045)	0.070 (0.049)
Data Subjects & Recipients	0.090 (0.034)	0.093 (0.060)	0.123 (0.052)	0.103 (0.052)
Retention & Deletion	0.218 (0.087)	0.183 (0.141)	0.283 (0.152)	0.229 (0.136)
Technical & Org. Measures	0.237 (0.140)	0.246 (0.158)	0.285 (0.100)	0.257 (0.135)

Table 4.14: Chapter-level METEOR mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.456 (0.120)	0.506 (0.136)	0.453 (0.124)	0.471 (0.128)
Purposes & Legal Bases	0.034 (0.037)	0.034 (0.047)	0.015 (0.019)	0.027 (0.037)
Personal Data Categories	0.079 (0.073)	0.140 (0.116)	0.101 (0.086)	0.106 (0.096)
Data Subjects & Recipients	0.199 (0.078)	0.203 (0.134)	0.270 (0.105)	0.225 (0.112)
Retention & Deletion	0.332 (0.114)	0.245 (0.160)	0.362 (0.177)	0.314 (0.160)
Technical & Org. Measures	0.295 (0.108)	0.280 (0.145)	0.305 (0.079)	0.293 (0.113)

Table 4.15: Chapter-level BERTScore Precision mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.855 (0.015)	0.862 (0.020)	0.855 (0.016)	0.857 (0.017)
Purposes & Legal Bases	0.560 (0.009)	0.570 (0.025)	0.568 (0.016)	0.566 (0.019)
Personal Data Categories	0.670 (0.022)	0.690 (0.040)	0.698 (0.032)	0.686 (0.034)
Data Subjects & Recipients	0.686 (0.018)	0.700 (0.035)	0.712 (0.030)	0.700 (0.030)
Retention & Deletion	0.814 (0.033)	0.817 (0.050)	0.840 (0.037)	0.824 (0.042)
Technical & Org. Measures	0.841 (0.038)	0.858 (0.031)	0.863 (0.027)	0.854 (0.033)

Table 4.16: Chapter-level BERTScore Recall mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.879 (0.019)	0.887 (0.025)	0.879 (0.022)	0.882 (0.022)
Purposes & Legal Bases	0.670 (0.016)	0.667 (0.017)	0.662 (0.016)	0.666 (0.016)
Personal Data Categories	0.850 (0.012)	0.849 (0.015)	0.847 (0.011)	0.849 (0.013)
Data Subjects & Recipients	0.851 (0.016)	0.838 (0.017)	0.836 (0.022)	0.842 (0.020)
Retention & Deletion	0.890 (0.018)	0.879 (0.023)	0.897 (0.027)	0.889 (0.024)
Technical & Org. Measures	0.889 (0.018)	0.882 (0.032)	0.893 (0.022)	0.888 (0.025)

Table 4.17: Chapter-level BERTScore F1 mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.867 (0.016)	0.875 (0.022)	0.867 (0.017)	0.869 (0.019)
Purposes & Legal Bases	0.610 (0.009)	0.614 (0.014)	0.611 (0.010)	0.612 (0.012)
Personal Data Categories	0.749 (0.016)	0.761 (0.028)	0.765 (0.022)	0.759 (0.023)
Data Subjects & Recipients	0.760 (0.014)	0.762 (0.024)	0.769 (0.022)	0.764 (0.020)
Retention & Deletion	0.850 (0.024)	0.847 (0.035)	0.867 (0.031)	0.855 (0.031)
Technical & Org. Measures	0.864 (0.027)	0.870 (0.027)	0.878 (0.019)	0.871 (0.025)

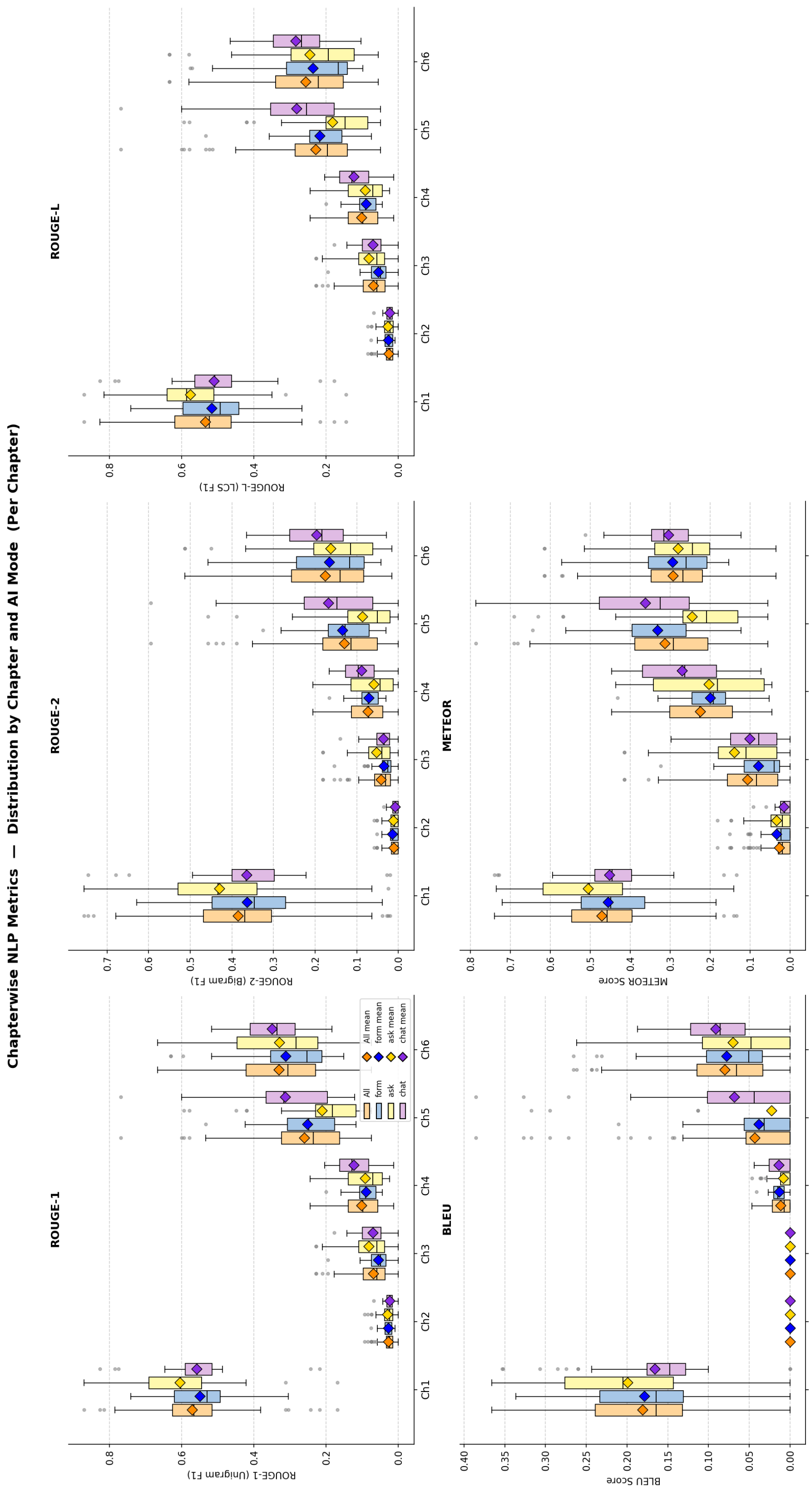

Figure 4.7: Chapter-level distributions of BLEU, ROUGE-1/2/L, and METEOR by mode (full sample $n = 113$).

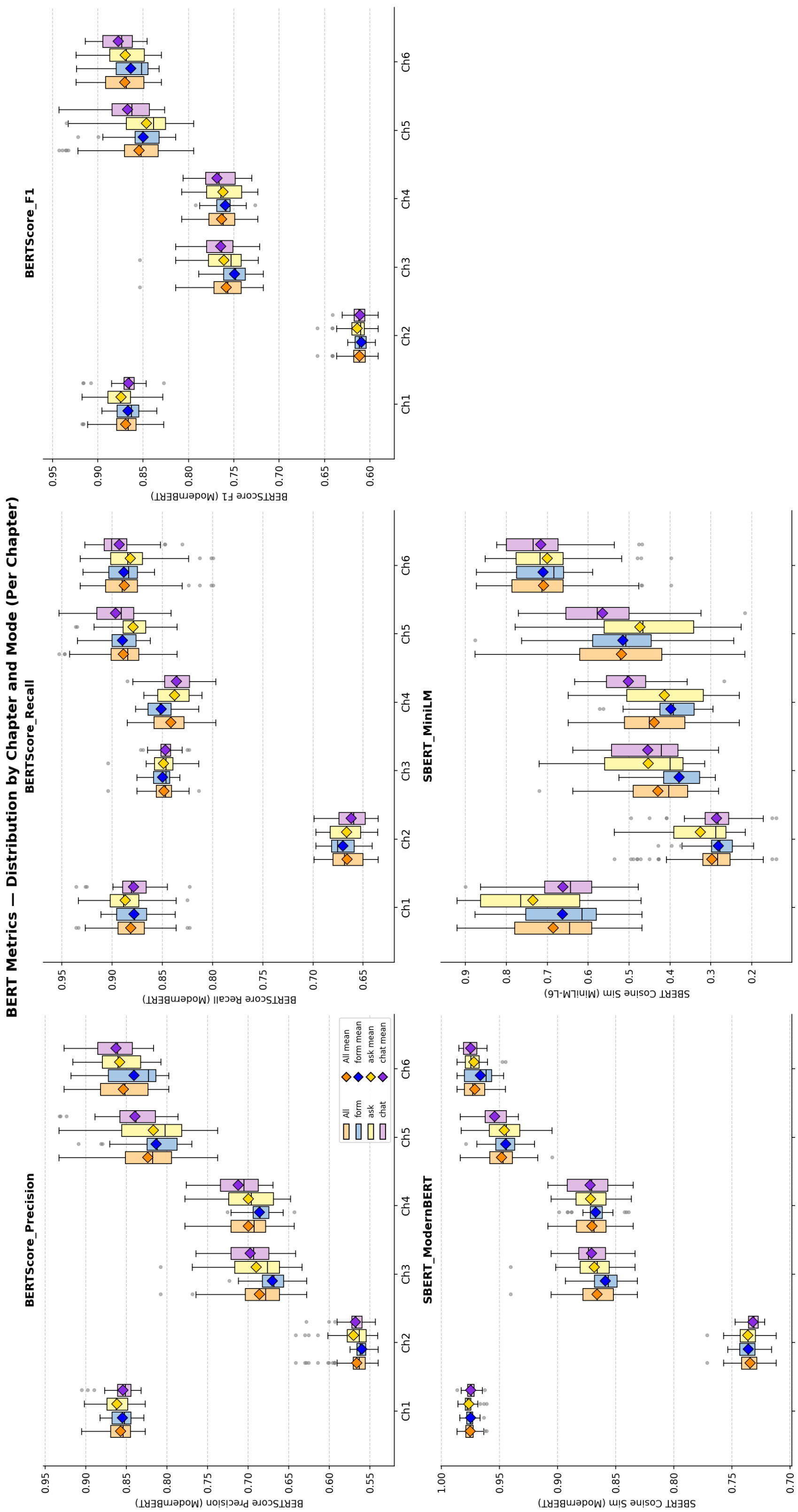

Figure 4.8: Chapter-level distributions of BERTScore Precision / Recall / F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$).

Table 4.18: Chapter-level SBERT (ModernBERT) cosine similarity mean (SD) by mode (full sample $n = 113$).

Chapter	Form	Ask	Chat	All
Controller & Contact	0.975 (0.004)	0.977 (0.006)	0.975 (0.005)	0.976 (0.005)
Purposes & Legal Bases	0.736 (0.009)	0.736 (0.012)	0.732 (0.007)	0.735 (0.010)
Personal Data Categories	0.859 (0.015)	0.869 (0.021)	0.871 (0.018)	0.866 (0.019)
Data Subjects & Recipients	0.868 (0.013)	0.872 (0.019)	0.872 (0.022)	0.871 (0.018)
Retention & Deletion	0.945 (0.013)	0.946 (0.018)	0.955 (0.014)	0.949 (0.016)
Technical & Org. Measures	0.967 (0.012)	0.972 (0.010)	0.975 (0.007)	0.971 (0.010)

Patterns across chapters. *Controller and Contact Details* contains the most templated content (company name, address, DPO contact); the per-mode spread is correspondingly small on every metric, and the absolute scores are the highest of the six chapters on the lexical metrics (BLEU ≈ 0.18, ROUGE-1 ≈ 0.57, METEOR ≈ 0.47). *Purposes and Legal Bases* is the weakest chapter across every metric, with BLEU collapsing to zero and METEOR below 0.04 in every mode; this chapter is dominated by free-text legal formulation, and the reference document and participant documents share almost no surface n-grams even where the underlying content overlaps. *Categories of Personal Data* and *Data Subjects and Recipients* sit in an intermediate range (0.06–0.27 on the lexical metrics) with Chat scoring highest on *Data Subjects* (METEOR 0.270 vs $0.199/0.203$ for Form/Ask). *Retention Periods* and *Technical and Organisational Measures* are the chapters where Chat establishes its clearest advantage: on Retention, Chat leads on every metric (BERTScore F1 0.867, BLEU 0.068, METEOR 0.362); on T&O Measures, Chat leads on BLEU (0.091), ROUGE-1 (0.350), BERTScore F1 (0.878), and SBERT (0.975).

4.5.3 Document-Level versus Chapter-Level Comparison

This subsection compares the two granularities and identifies where chapter-level scoring tells a different story than document-level. A direct comparison of the document-level and chapter-level means is provided by Table 4.19. Document-level means are substantially higher than the cross-chapter means for every metric: for example, the aggregated document-level BERTScore F1 of 0.902 exceeds the average chapter-level F1 of 0.788 by approximately 0.114. The same gap is observable for the lexical metrics, and most pronounced for ROUGE-1 ($\Delta \approx 0.26$). This is consistent with the document-level calculation pooling token overlap across all six chapters, which dilutes chapter-specific zero-overlap regions (notably the Purposes and Legal Bases chapter) and pushes the aggregate scores upward.

The reference-based metrics therefore neither confirm nor cleanly contradict the Form-first ordering of the user-experience chapters (§4.1–§4.4): Form leads on recall-oriented and bigram-sensitive measures, Chat leads on embedding-based ones, and Ask trails throughout. Whether a reference-free, GDPR-specific quality signal sees the same split is the question taken up next in §4.6.

4.6 Document Quality — LLM-as-a-Judge

Building on the reference-based NLP metrics reported in §4.5, this section turns to a reference-free quality signal. The previous section measured similarity to a single expert-authored Form-mode reference; here, the judge evaluates each document against a rubric rather than a target, which lifts the implicit advantage that the reference document conferred on Form-style outputs. The LLM-as-a-Judge evaluation provides a reference-free, GDPR-specific quality signal across five dimensions: *Completeness*, *Scenario Faithfulness*, *Legal Correctness*, *Hallucination* (inverse-coded; higher equals less hallucination), and *Overall*. The dimension-wise rubric design follows the precedent established by Su et al.'s JuDGE benchmark for LLM-evaluated legal-document generation [15]. The judge is a single LLM (Gemini 3.1 Pro Preview, temperature 0) prompted with the rubric documented in Appendix C.1; the resulting scores are reported as a supplementary, single-pass single-model signal alongside the survey instruments and reference-based NLP metrics, and without inter-rater replication or legal-expert validation they should be read as indicative rather than definitive. All 113 retained documents are scored. Per-mode results are reported in the order Form, Ask, Chat, followed by the aggregated mean across all documents.

4.6.1 Per-Dimension Results

This subsection walks through each rubric dimension in turn, reporting per-mode means, medians, and dispersion before the holistic Overall score. The five rubric

Table 4.19: Document-level versus mean chapter-level metric values by mode. The mean chapter-level value is the unweighted average across the six chapters. Differences are document-level minus chapter-level mean. Source: `doc_vs_chapter_diff.csv`.

Metric	Form		Ask		Chat	
	Doc	Chapter	Doc	Chapter	Doc	Chapter
BLEU	0.129	0.051	0.127	0.050	0.143	0.057
ROUGE-1	0.488	0.215	0.465	0.226	0.511	0.240
ROUGE-2	0.306	0.131	0.257	0.134	0.291	0.144
ROUGE-L	0.388	0.191	0.347	0.202	0.389	0.216
METEOR	0.412	0.232	0.365	0.235	0.390	0.251
BERTScore Precision	0.889	0.738	0.886	0.750	0.895	0.756
BERTScore Recall	0.923	0.838	0.908	0.834	0.913	0.836
BERTScore F1	0.905	0.783	0.897	0.788	0.904	0.793
SBERT (ModernBERT)	0.986	0.892	0.986	0.895	0.990	0.897

dimensions are reported in the order Completeness $\rightarrow$ Faithfulness $\rightarrow$ Legal Correctness $\rightarrow$ Hallucination $\rightarrow$ Overall to allow the picture to build cumulatively before the holistic Overall score is read. Table 4.20 reports the per-dimension mean and standard deviation per mode and the aggregated mean. Figure 4.9 shows the corresponding distributions.

Table 4.20: LLM-as-Judge dimension means $\pm$ standard deviation by mode (full sample $n = 113$, scale 1–5, higher is better including for *Hallucination (inverse)*). The aggregated "All" column is the mean across all 113 documents.

Dimension	Form	Ask	Chat	All
Completeness	4.84 ± 0.55	3.84 ± 1.54	3.18 ± 1.59	3.94 ± 1.43
Scenario Faithfulness	1.84 ± 0.76	2.30 ± 1.20	2.49 ± 1.19	2.21 ± 1.10
Legal Correctness	3.59 ± 0.55	3.43 ± 1.34	3.15 ± 1.42	3.39 ± 1.18
Hallucination (inverse)	1.65 ± 0.72	2.40 ± 1.12	2.54 ± 0.91	2.20 ± 1.02
Overall	2.41 ± 0.69	2.49 ± 1.04	2.62 ± 1.31	2.50 ± 1.04

As with BERTScore Recall (§4.5), Form leads strongly on Completeness. For *Completeness*, Form scores highest by a substantial margin (mean 4.84, median 5.0), with Ask second (3.84, median 5.0) and Chat lowest (3.18, median 3.0); aggregated mean 3.94. Completeness decreases by approximately 1.66 points from Form to Chat. Form's standard deviation is also the lowest of the three modes (0.55 versus Ask 1.54 and Chat 1.59), indicating that Form yields uniformly complete documents.

The picture changes when faithfulness to the specific study scenario is asked about. For *Scenario Faithfulness*, the ordering inverts. Form scores lowest (1.84, median 2.0), Ask second (2.30, median 2.0), Chat highest (2.49, median 2.0); aggregated mean 2.21. Faithfulness to the specific study scenario increases by approximately 0.65 points from Form to Chat.

Legal Correctness restores the Completeness ordering, but on a much narrower scale. For *Legal Correctness*, Form scores highest (3.59, median 4.0), Ask second (3.43, median 4.0), Chat lowest (3.15, median 3.0); aggregated mean 3.39. The spread is approximately 0.45 points across modes; Form's standard deviation (0.55) is again the lowest, reflecting a narrow concentration of Legal Correctness scores around the value 4.0.

Hallucination tracks Faithfulness almost exactly. For *Hallucination (inverse)*, the or-

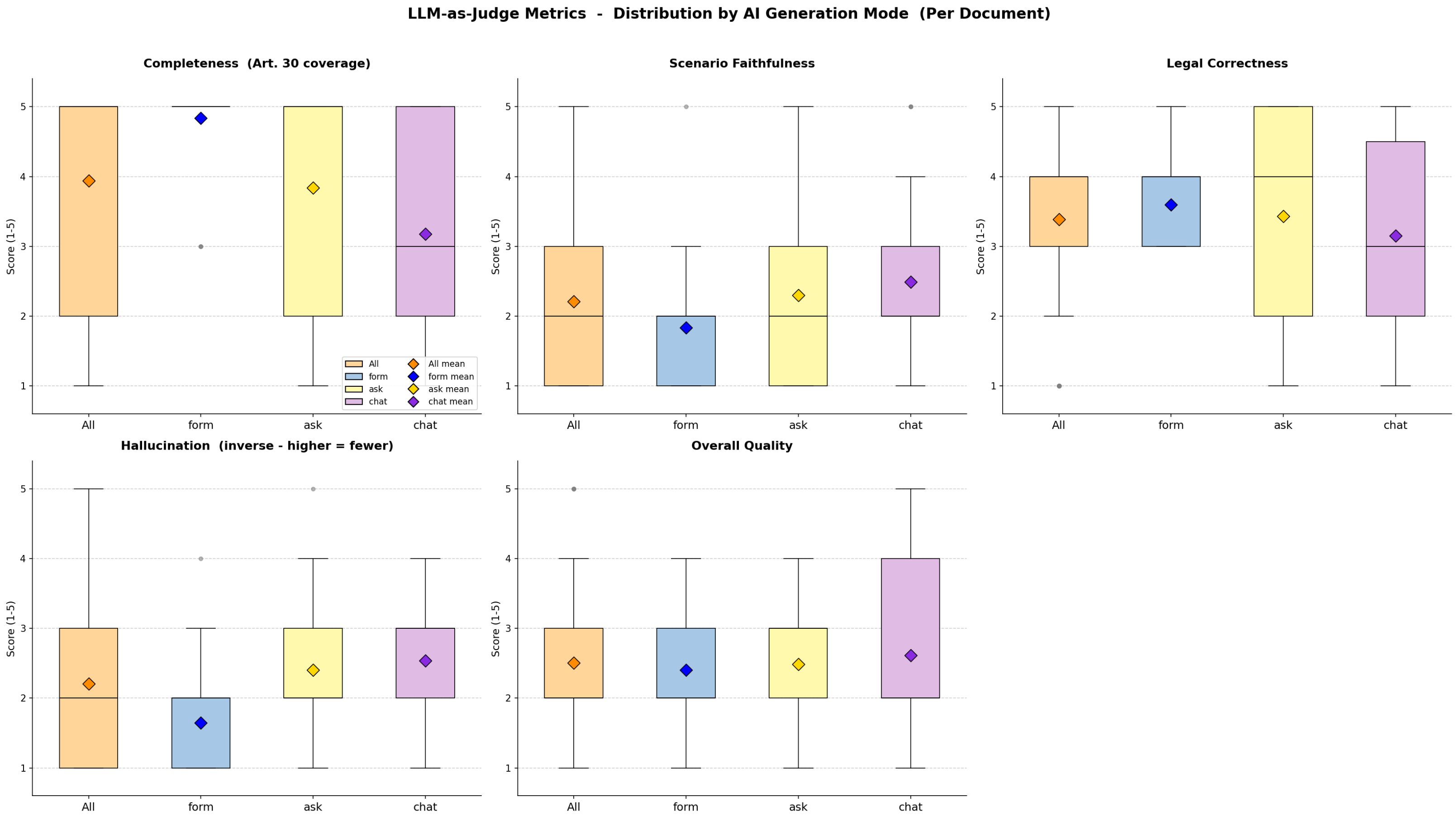

Figure 4.9: Distributions of the five LLM-as-Judge dimensions by mode (full sample $n = 113$, scale 1–5).

dering matches Scenario Faithfulness. Form scores lowest (1.65, median 2.0), Ask second (2.40, median 2.0), Chat highest (2.54, median 3.0); aggregated mean 2.20. Because the dimension is inverse-coded, this means Form documents are judged to contain the most fabricated content and Chat documents the least. Combining this dimension with Completeness reported above, 31 of the 37 Form documents ($83.8\,\%$) achieve Completeness $= 5$ while simultaneously scoring Hallucination (inverse) ≤ 2 ("moderate" or "significant" hallucination per the rubric), and 17 of these combine the maximum Completeness score with the minimum Hallucination (inverse) score of 1; this co-occurrence is the proximate source of the Form-versus-Faithfulness inversion drawn out in §4.6.2 below.

The Overall dimension compresses the four contrasting orderings above into a single judgement. For *Overall*, the three modes are clustered closely together: Form 2.41, Ask 2.49, Chat 2.62; aggregated 2.50. The aggregate Overall score thus sits below the midpoint of the 1–5 scale across all modes. Form's standard deviation (0.69) is again the lowest, while Chat's is the highest (1.31), indicating that Chat produces both the highest- and lowest-rated documents in the sample.

4.6.2 Patterns Across the Five Dimensions

This subsection steps back from the individual rubric dimensions and draws out the contrasts between them across modes. Three patterns emerge from the per-dimension means. First, the Completeness ordering (Form $>$ Ask $>$ Chat) is the inverse of the Scenario Faithfulness and Hallucination orderings (Form $<$ Ask $<$ Chat). Form documents are judged to be the most structurally complete but also the most prone to fabricated content, while Chat documents are judged to be the most faithful to the specific scenario but the least structurally complete. Ask sits between the two extremes on every dimension.

Second, the Legal Correctness scores are flat relative to the Completeness spread: the gap between modes is approximately 0.45 points on Legal Correctness compared to 1.66 points on Completeness. The judge therefore distinguishes between modes more sharply on whether the document covers all required Article 30(1) fields than on whether the legal terminology is appropriate.

Third, the Overall scores compress the per-dimension spread into a narrow range

(Form 2.41 to Chat 2.62, $\Delta = 0.21$), with the ordering Form $<$ Ask $<$ Chat. This is consistent with Overall reflecting a holistic judgement that weighs Faithfulness and Hallucination against Completeness; Form's high Completeness is partially offset by its low Faithfulness and high Hallucination, and the net result is a slight Chat advantage. The interpretation of the Form Completeness–Hallucination co-occurrence reported above in terms of the checkbox interface design is reserved for the Discussion chapter.

The judge data thus complete the document-quality picture begun in §4.5: Form leads on Completeness and Legal Correctness, Chat leads on Faithfulness and Hallucination, and the holistic Overall score collapses these tensions into a narrow range. The participant-level demographics that frame the entire sample are reported finally in §4.8.

4.7 Summary of Findings

This section consolidates the descriptive and inferential results of the preceding evaluation, organised by measurement family. Three relative orderings recur across measurement families. Form leads on usability, workload, first-place mode preference, and the lexical reference-based NLP metrics (BLEU, ROUGE family, METEOR, and BERTScore Recall). Chat leads on the semantic similarity metrics (BERTScore F1, SBERT cosine) and on Scenario Faithfulness in the LLM-as-Judge rubric. Ask sits in the middle on most measures and is descriptively last on every reference-based NLP metric. The numerical detail is consolidated below.

Usability (SUS). Form yields the highest aggregate SUS-item ratings on the positively worded items (item 3 "easy to use": Form 4.38, Ask 4.01, Chat 3.99; item 7 "learn quickly": Form 4.30, Ask 4.00, Chat 4.03) and the lowest aggregate ratings on the negatively worded items (item 8 "cumbersome": Form 1.65, Ask 2.16, Chat 2.29). Within- system confidence (item 9) is essentially flat across modes (3.83/3.84/3.90). The aggregated Friedman test on the per-mode SUS aggregate is non-significant ($\chi^2(2) = 5.51$, $p = 0.064$, $W = 0.040$); Bonferroni-corrected Wilcoxon post-hoc tests yield a significant Form-versus-Ask difference ($W = 578.5$, $p = 0.013$, $r = +0.368$), a Form-versus-Chat difference at the threshold ($p = 0.018$), and no Ask-versus-Chat difference ($p = 0.666$). Form leads on the positively keyed items and on the cumbersomeness item; Ask and Chat overlap closely; within-system confidence (item 9) is flat across modes.

Workload (NASA R-TLX). Form returns the lowest mean on every demand subscale: Mental Demand 5.77 vs. 6.80/6.36, Physical Demand 2.77 vs. 3.54/3.26, Temporal Demand 4.16 vs. 4.97/5.54, Effort 5.68 vs. 6.39/6.32, and Frustration 5.88 vs. 7.58/7.13 for Ask and Chat respectively. Perceived self-Performance is essentially flat (8.71/8.68/9.39). The aggregated Friedman test across the six subscales is non-significant ($\chi^2(2) = 5.24$, $p = 0.073$, $W = 0.038$). Two subscales reach significance individually: Temporal Demand ($\chi^2(2) = 12.77$, $p = 0.002$, $W = 0.093$) and Frustration ($\chi^2(2) = 8.93$, $p = 0.012$, $W = 0.065$). Significant pairwise contrasts are confined to Form versus the LLM-driven modes, with medium-to-large rank-biserial effects ($|r| = 0.36$ to 0.57); Ask versus Chat is non-significant on every subscale. Form returns the lowest mean on every demand subscale; perceived self-Performance is essentially flat across modes.

Perceived AI Support. The six AI-support items pooled together are flat across modes (aggregated Friedman test $\chi^2(2) = 1.05$, $p = 0.592$, $W = 0.008$). At the item level, the legal-basis confidence item (5_{ai}) increases monotonically with LLM involvement (Form 3.58, Ask 3.74, Chat 3.81; aggregated 3.71; aggregated Friedman test non-significant, $\chi^2(2) = 3.76$, $p = 0.152$), while the re-use intent item (6_{ai}) reverses (Form 3.99, Ask 3.55, Chat 3.42; aggregated 3.65; aggregated Friedman test significant, $\chi^2(2) = 11.12$, $p = 0.004$, $W = 0.081$; Form versus Chat $r = +0.543$, large effect). Transparency (3_{ai}) is highest in Ask (4.16 vs. Form 3.51 and Chat 3.75). All three modes are essentially flat on the AI-helpfulness ratings; re-use intent reverses with LLM involvement.

Mode Preference. On the post-experiment ranking ($n = 69$), Form receives $53.6\,\%$ of first-place votes (37 of 69), Chat $29.0\,\%$ (20), and Ask $17.4\,\%$ (12); mean ranks are 1.78, 2.07, and 2.14. A chi-square goodness-of-fit test against a uniform distribution is significant ($\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$, medium effect). The Friedman test on the full rank distribution is non-significant

($\chi^2(2) = 5.07$, $p = 0.079$). Form receives the largest first-place share ($53.6\,\%$); Chat is second; Ask trails.

NLP Metrics (document level, $n = 113$). Means by mode (Form / Ask / Chat / aggregate): BLEU 0.129 / 0.127 / 0.143 / 0.133; ROUGE-1 0.488 / 0.465 / 0.511 / 0.488; ROUGE-2 0.306 / 0.257 / 0.291 / 0.285; ROUGE-L 0.388 / 0.347 / 0.389 / 0.375; METEOR 0.412 / 0.365 / 0.390 / 0.389; BERTScore Precision 0.889 / 0.886 / 0.895 / 0.890; BERTScore Recall 0.923 / 0.908 / 0.913 / 0.915; BERTScore F1 0.905 / 0.897 / 0.904 / 0.902; SBERT cosine similarity (ModernBERT) 0.986 / 0.986 / 0.990 / 0.987. Aggregated Friedman tests are significant for every metric: BLEU ($\chi^2(2) = 6.23$, $p = 0.044$), ROUGE-1 (10.69, $p = 0.005$), ROUGE-2 (13.46, $p = 0.001$), ROUGE-L (11.31, $p = 0.004$), METEOR (9.92, $p = 0.007$), BERTScore Precision (18.54, $p < 0.001$), BERTScore Recall (30.77, $p < 0.001$, $W = 0.410$, the largest effect in the pipeline), BERTScore F1 (13.00, $p = 0.002$), and SBERT (16.69, $p < 0.001$). Across pairwise post-hoc tests, Form dominates on BERTScore Recall (Form vs. Ask $r = +1.000$; Form vs. Chat $r = +0.687$) and on ROUGE-2 (Form vs. Ask $r = +0.829$), while Chat dominates on SBERT (Chat vs. Form and Chat vs. Ask significant with large effects). Ask underperforms both other modes on every metric. Form leads on the lexical metrics (BLEU, ROUGE family); Chat leads on the semantic metrics (BERTScore F1, SBERT); Ask is descriptively last on every reference-based metric.

LLM-as-a-Judge (1–5 scale). Means by mode (Form / Ask / Chat / aggregate): Completeness 4.84 / 3.84 / 3.18 / 3.94; Scenario Faithfulness 1.84 / 2.30 / 2.49 / 2.21; Legal Correctness 3.59 / 3.43 / 3.15 / 3.39; Hallucination (inverse-coded; higher equals less hallucination) 1.65 / 2.40 / 2.54 / 2.20; Overall 2.41 / 2.49 / 2.62 / 2.50. Form has the lowest standard deviation on Completeness (0.55 vs. $1.54/1.59$), Legal Correctness (0.55), and Overall (0.69). On the joint distribution, 31 of 37 Form documents ($83.8\,\%$) score Completeness $= 5$ together with Hallucination (inverse) ≤ 2; 17 Form documents combine the maximum Completeness score of 5 with the minimum Hallucination (inverse) score of 1. Form leads on Completeness; Chat leads on Scenario Faithfulness; the Form Completeness $= 5$ and low-Hallucination co-occurrence ($83.8\,\%$) is the proximate source of the Form-versus-Faithfulness inversion.

The interpretation of these patterns against the research questions is taken up in Chapter 5.

4.8 Participant Demographics

The preceding sections reported the user-experience and document-quality findings without distinguishing among participants. This closing section describes the demographic profile of the matched within-subjects sample of $n = 69$ valid participants ($p_{0001} \geq 1$) on which all of those findings rest. Four screened records were excluded from all subsequent analyses for incomplete or invalid participation, yielding the analysis sample reported in Chapter 3. The profile is presented descriptively here; its implications for the generalisability of the findings are reserved for the Discussion chapter.

4.8.1 Numeric and Ordinal Variables

Table 4.21 reports the descriptive statistics for the four numeric or ordinal variables collected in the demographics block: total session duration in seconds, and three five-point Likert self-reports of prior expertise (GDPR knowledge, ROPA familiarity, chatbot familiarity).

Table 4.21: Descriptive statistics for numeric and ordinal demographic variables ($n = 69$). Duration is in seconds and aggregates the full study session across all three modes; the three expertise variables are five-point Likert self-reports (1 = no familiarity, 5 = expert). The duration minimum of -1 is a sentinel value for one non-responding record on this single variable; the median is a more robust point estimate.

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
Duration (s)	69	1865.7	1354.2	-1	840	2013	2708	5788
GDPR knowledge	69	2.81	1.15	1	2	3	4	5
ROPA familiarity	69	1.45	0.76	1	1	1	2	4
Chatbot familiarity	69	4.36	0.73	2	4	4	5	5

The median session duration is 2013 s (≈ 33.6 min) per participant across the full session, consistent with the protocol target of ≈ 90 min total once spread across three modes and the rating instruments. ROPA familiarity is the lowest of the four variables (mean 1.45, median 1), confirming that the sample qualifies as ROPA-novice in line with the inclusion criteria. Chatbot familiarity is correspondingly high (mean 4.36, median 4). GDPR knowledge sits near the scale midpoint (mean 2.81).

4.8.2 Categorical Variables

Table 4.22 reports counts and shares for the five categorical demographic variables: gender, age (binned by year), highest completed degree, current professional status, and field of study or work. Three participants did not report gender; the share columns therefore use the per-variable response base.

Table 4.22: Categorical demographic variables ($n = 69$ unless noted). Shares are computed within variable.

Variable	Level	n	Share
Gender ($n = 66$ reporters)	Male	41	62.1 %
	Female	24	36.4 %
	Other	1	1.5 %
Age (years)	24	18	26.1 %
	25	18	26.1 %
	26	10	14.5 %
	Other (21–40)	23	33.3 %
Degree	Bachelor	60	88.2 %
	Master	6	8.8 %
	Other	2	2.9 %
Profession	Student	56	81.2 %
	Employed	10	14.5 %
	PhD	2	2.9 %
	Staatsexamen	1	1.5 %
Field	IT	41	59.4 %
	Business (BWL)	16	23.2 %
	Other	6	8.7 %
	Research	4	5.8 %
	HR	2	2.9 %

The sample is dominated by Bachelor-level students of IT or Computer Science. Of the 66 participants who reported gender, $62.1\,\%$ identified as male. The full distribution of demographics and expertise variables is visualised in Figure 4.10. This profile — IT-leaning, chatbot-fluent, ROPA-novice — frames the results reported in §4.1–§4.6 and is taken up again in the Discussion chapter when generalisability is addressed.

Figure 4.10: Demographic and expertise distributions: histograms of age and session duration, count plots for gender, degree, profession, and field, and a grouped bar chart of the three expertise self-reports (GDPR, ROPA, chatbot) on the five-point Likert scale.

5 Discussion

This chapter interprets the descriptive and inferential patterns reported in Chapter 4 against the two research questions of this thesis. The numerical results themselves are not restated here; the reader is referred to Section 4.7 for the consolidated recap.

5.1 RQ1: User Experience Across Modes

On the self-report instruments, the structured Form mode is rated the most usable, the least demanding, and the most preferred. Form receives the largest share of first-place preference votes ($53.6\,\%$), the highest aggregate SUS-item ratings, and the lowest means on every R-TLX demand subscale. The Friedman omnibus tests on the SUS and R-TLX measures do not reach the conventional significance threshold; the significant sub-effects on the matched-pairs subset are confined to Form versus the LLM-driven modes on Temporal Demand, Frustration, and re-use intent.

A pattern that recurs across every aggregate user-experience measure is that *Ask and Chat are descriptively and inferentially indistinguishable* from each other: pairwise tests between the two LLM-driven modes are non-significant on every SUS aggregate, every R-TLX subscale, every pooled AI-support item, and the rank distribution. The user-experience contrast in this study is therefore best read as a binary one between the structured questionnaire (Form) and the LLM-driven interactions (Ask, Chat together), rather than as the three-way ordering implied by the original “more LLM = better” framing.

The Perceived AI Support items do show one monotonic pattern with LLM involvement: legal-basis confidence (item 5_{ai}) rises from Form to Chat. The same items show a counter-pattern on re-use intent (item 6_{ai}), where Form is rated highest. The free-text comments characterise Form as fast and ergonomic, Ask as transparent but cumbersome, and Chat as comfortable but opaque, which is consistent with the quantitative split.

5.2 RQ2: Document Quality and the Confidence Gap

The document-quality side of the study yields a more nuanced ordering than the self-report side. Reference-based NLP metrics place Form first on lexical recall (BERTScore Recall, ROUGE-2) and the n-gram- based metrics, Chat first on semantic similarity (SBERT) and BLEU, and Ask consistently last on every metric. The chapter-level breakdown attributes Chat's BLEU and METEOR advantage primarily to the Retention Periods and the Technical and Organisational Measures chapters.

The LLM-as-Judge inverts the BERTScore Recall ordering. Form leads on Completeness (4.84) but lags on Scenario Faithfulness (1.84) and Hallucination (inverse, 1.65); Chat leads on Faithfulness and (inverse) Hallucination but trails on Completeness; Overall scores are clustered tightly between 2.41 and 2.62. The proximate source of this inversion is the joint distribution noted in Section 4.7: $83.8\,\%$ of Form documents combine maximum Completeness with moderate-to-significant hallucination, an artefact of the checkbox interface that produces structurally complete but scenario-detached documents. This mixed picture sits uneasily with the implicit "more LLM equals better quality" assumption that motivates the architecturally elaborated pipelines of recent legal-document generation work [10, 9, 7].

Reading RQ1 and RQ2 together surfaces an apparent gap between *perceived* AI support and *objective* document quality. Users rate Form as the most usable and prefer it most, while the mode that most often matches the reference at the semantic level (Chat) is rated as opaque and ranked second. The mode that scores highest on AI-mediated legal-basis confidence (Chat) is also the mode that produces the documents the judge rates as least complete. This parallels Surya et al.'s evaluation of an interactive legal AI assistant [16], where ease of use was the headline finding — a pattern that here translates into the structured Form being preferred despite its weaker faithfulness profile. The implications of this gap for the design of LLM-supported ROPA tooling, and the extent to which the present novice sample generalises, are addressed in the remainder of this chapter.

6 Conclusion

A Study Scenario

The scenario reproduced below was presented to all participants in German, regardless of interaction mode (Form, Ask, Chat). The English translation is provided for the international reader; the German original is the version actually shown.

A.1 Original (German)

```
Szenario: Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung

Sie sind Mitarbeitender namens A.I. und arbeiten für die APOR GmbH, ein
Softwareunternehmen mit ca. 50 Mitarbeitenden in Ulm. Sie handeln im Namen
des Unternehmens und sind im Szenario sowohl Verantwortlicher (Data
Controller) als auch Datenschutzbeauftragte:r (DPO/DSB).

Zur Organisation:

Rolle: Data Controller, DPO
Adresse: Mainstraße 67, Ulm
E-Mail: mail@apor.eu
Telefon: 012345678
Vertreter: (Sie), A. I., ai@apor.eu
DPO: Das sind auch Sie

Verarbeitung
Das Unternehmen verarbeitet personenbezogene Daten, um Mitarbeitende vom
Eintritt bis zum Austritt zu verwalten. Im Fokus stehen:
```

- Anlegen neuer Mitarbeitender
- Dokumentation von Arbeitszeiten
- Verwaltung von Urlaub und Abwesenheiten

Zusätzlich verwaltet APOR für berechtigte Mitarbeitende eine Parkplatzberechtigung für einen Firmenparkplatz mit Schranke. Damit berechtigte Mitarbeitende den Parkplatz nutzen können, wird ein geeignetes Merkmal zur Zuordnung der Berechtigung hinterlegt (z. B. ein Fahrzeugkennzeichen).

APOR arbeitet mit intern gehostetem Microsoft Office sowie internen Servern (keine externe Cloud). Krankentage oder medizinische Angaben sind für diese Verarbeitung nicht erforderlich.

Für die Verwaltung sind technische und organisatorische Maßnahmen vorgesehen, wie z.B. Zugriffsbeschränkung nach Rollen (Need-to-know), Passwortschutz (ggf. MFA), regelmäßige Backups, Zutrittskontrolle zu Serverraum/Archiv und Vertraulichkeitsverpflichtung der berechtigten Mitarbeitenden.

Alle in diesem Zusammenhang gespeicherten Informationen werden spätestens 3 Monate nach Austritt der jeweiligen Person gelöscht.

A.2 English Translation

Scenario: Employee Administration (Working Time / Leave) & Parking Authorisation

You are an employee named A.I., working for APOR GmbH, a software company with approximately 50 employees in Ulm. You act on behalf of the company and, in this scenario, are both the Data Controller and the Data Protection Officer (DPO).

Organisation details:

Role: Data Controller, DPO
Address: Mainstrasse 67, Ulm
E-mail: mail@apor.eu
Phone: 012345678
Representative: (You), A. I., ai@apor.eu
DPO: Also you

Processing
The company processes personal data in order to administer employees from onboarding to exit. The focus is on:

- Onboarding new employees
- Documentation of working hours
- Administration of leave and absences

In addition, APOR manages a parking authorisation for entitled employees for a company car park with a barrier. To allow entitled employees to use the car park, a suitable feature for assigning the authorisation is recorded (e.g. a vehicle licence plate).

APOR uses internally hosted Microsoft Office and internal servers (no external cloud). Sick days or medical information are not required for this processing.

For administration, technical and organisational measures (TOMs) are provided, such as role-based access restriction (need-to-know), password protection (where appropriate, MFA), regular backups, physical access control to the server room / archive, and a confidentiality obligation for authorised employees.

All information stored in this context is deleted no later than 3 months after the person's departure.

B Survey Instruments

This appendix reproduces the verbatim items of the two standardised survey instruments referenced from the Methodology chapter: the System Usability Scale (SUS) and the Raw NASA Task Load Index (R-TLX). The six-item Perceived AI Support instrument is bespoke to this study and is listed in full in the Methodology chapter; it is not reproduced here.

B.1 System Usability Scale (SUS)

Five-point Likert scale, $1 =$ Strongly Disagree, $5 =$ Strongly Agree. Odd-numbered items are positively keyed; even-numbered items are negatively keyed. The canonical wording reproduced below is taken from Brooke [3].

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

B.2 NASA Raw Task Load Index Subscales

Six subscales, each rated on the standard NASA TLX 21-point scale. The Raw TLX variant used in this study takes the unweighted mean across the six subscales rather than applying the pairwise-comparison weighting of the original NASA TLX. The subscale definitions reproduced below follow Hart [6].

Mental Demand How mentally demanding was the task?

Physical Demand How physically demanding was the task?

Temporal Demand How hurried or rushed was the pace of the task?

Performance How successful were you in accomplishing what you were asked to do?

Effort How hard did you have to work to accomplish your level of performance?

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

C LLM-as-a-Judge Configuration

This appendix documents the inputs to the LLM-as-a-judge evaluation layer that scored the 113 retained ROPA documents. It contains the verbatim system and user prompts, the five-dimension rubric, and the model configuration needed to reproduce the evaluation.

C.1 System and User Prompt

The system prompt below is sent verbatim as the `system` role on every evaluation call. It is taken from `python/judge/prompt.json` (key `judge.system`).

```
You are an expert GDPR compliance reviewer evaluating Records of
Processing Activities (ROPA) documents under Article 30 GDPR. The
documents are written by novices (non-experts, no prior GDPR training)
who described the same processing scenario using different LLM-assisted
tools. Your job is to judge each ROPA document against (a) the provided
scenario and (b) GDPR Article 30 requirements. Be strict but fair. Do not
reward verbosity. Do not penalise a document for merely restating
scenario facts. Respond with valid JSON only - no prose, no Markdown
fences, no commentary before or after the JSON object.
```

The accompanying user-message template is sent on every call with `{scenario}` (the German original from Appendix A.1) and `{document}` (the participant ROPA document) substituted in:

```
You will evaluate one ROPA document against the scenario it is supposed
```

```
to cover.

=== SCENARIO (verbatim, German) ===
{scenario}
=== END SCENARIO ===

=== ROPA DOCUMENT TO EVALUATE ===
{document}
=== END DOCUMENT ===

Score the document on each of the following five metrics using a 1-5
Likert scale, and give a short rationale (max 2 sentences, English) for
each score. The scale for all metrics is:
1 = very poor, 2 = poor, 3 = adequate, 4 = good, 5 = excellent.

Metrics and what to assess:
- completeness: Coverage of Art. 30(1) GDPR required elements -
  controller identity and contact, DPO, purposes of processing, legal
  basis, categories of data subjects and personal data, categories of
  recipients, retention periods, and technical/organisational measures
  (TOMs). Missing elements lower the score.
- scenario_faithfulness: How accurately the document reflects the
  concrete facts of THIS scenario - APOR GmbH, Ulm (Mainstrasse 67), ~50
  employees, internally hosted Microsoft Office and internal servers (NO
  external cloud), no medical/sickness data, 3-month deletion after
  employee departure, parking lot authorisation using a distinguishing
  feature such as a licence plate. Generic ROPA boilerplate that ignores
  these specifics lowers the score.
- legal_correctness: Correct use of GDPR terminology and correct citation
  of legal basis (typically Art. 6(1)(b) performance of contract for
  employment data, Art. 6(1)(c) legal obligation where applicable, Art.
  6(1)(f) legitimate interest for parking authorisation). Wrong articles,
  invented legal bases, or confusing controller/processor/DPO roles lower
  the score.
- hallucination_inverse: Inverse score for invented facts not supported
```

```
  by the scenario (e.g. invented third-party processors, invented data
  categories like medical data, invented addresses, invented retention
  periods different from 3 months). 5 = no hallucinations, 1 = many
  invented facts.
- overall: Holistic judgement of how useful this document would be as a
  real Art. 30 ROPA record for this scenario, taking the four dimensions
  above into account.

Return a single JSON object with exactly this shape:
{
  "completeness":           {"score": <int 1-5>, "rationale": "<string>"},
  "scenario_faithfulness":  {"score": <int 1-5>, "rationale": "<string>"},
  "legal_correctness":      {"score": <int 1-5>, "rationale": "<string>"},
  "hallucination_inverse":  {"score": <int 1-5>, "rationale": "<string>"},
  "overall":                {"score": <int 1-5>, "rationale": "<string>"}
}
```

C.2 Rubric — Five Dimensions

All five dimensions share the same 1–5 Likert anchor: $1 =$ very poor, $2 =$ poor, $3 =$ adequate, $4 =$ good, $5 =$ excellent. The fourth dimension, `hallucination_inverse`, is inverted internally so that for all five dimensions higher equals better. For each dimension the judge returns both an integer score and a short English rationale (maximum two sentences).

C.3 Model Configuration and Limitations

The configuration in Table C.2 is taken from the `judge` block of `python/judge/prompt.json` and the corresponding cells of `ROPAgen_JUDGE.ipynb`. It is the minimal information required to reproduce the evaluation layer.

Dimension	What is assessed
Completeness	Coverage of Art. 30(1) GDPR required elements: controller identity and contact, DPO, purposes of processing, legal basis, categories of data subjects and personal data, categories of recipients, retention periods, and technical/organisational measures (TOMs). Missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of this scenario: APOR GmbH, Ulm (Mainstraße 67), ∼50 employees, internally hosted Microsoft Office and internal servers (no external cloud), no medical/sickness data, 3-month deletion after employee departure, parking authorisation using a distinguishing feature such as a licence plate. Generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis (typically Art. 6(1)(b) for employment data, Art. 6(1)(c) where applicable, Art. 6(1)(f) for parking authorisation). Wrong articles, invented legal bases, or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Inverse score for invented facts not supported by the scenario (e.g. invented third-party processors, invented categories such as medical data, invented addresses, retention periods other than 3 months). 5 = no hallucinations, 1 = many invented facts.
Overall	Holistic judgement of how useful the document would be as a real Art. 30 ROPA record for this scenario, taking the four dimensions above into account.

Table C.1: The five judge dimensions. All share the 1–5 Likert anchor; higher is better for every dimension after the hallucination inversion.

Setting	Value
Model	`google/gemini-3.1-pro-preview` (via OpenRouter)
Temperature	`0.0` (deterministic decoding)
Output format	JSON object, enforced via OpenRouter `response_format = json_schema` with `strict: true`
Schema name	`ropa_judge_result`
Required keys	`completeness`, `scenario_faithfulness`, `legal_correctness`, `hallucination_inverse`, `overall`
Per-key shape	`{"score": int in [1,5], "rationale": string}`
Additional properties	Disallowed at every object level
Documents evaluated	113 ROPA documents (Form 37, Ask 37, Chat 39)
Caching	Per-document JSON result cached on disk; reused on re-runs
Retry behaviour	Schema-validation or transient API failures retried; persistent failures logged and skipped
Logging	Each call logs model, latency, prompt and completion token counts, and raw response payload

Table C.2: LLM-as-a-judge model configuration.

Known limitations. The following limitations of the judge layer are stated for full transparency and discussed further in the Discussion chapter:

- **Single-judge bias.** Only one judge model (Gemini 3.1 Pro Preview) operationalises the rubric; inter-rater agreement with a human GDPR expert is not established.
- **Model-family bias.** Documents are produced by Mistral Large 3 inside ropa-gen, while the judge belongs to a different model family (Google). This reduces but does not eliminate stylistic-preference bias.
- **Single-scenario bias.** All scores are conditional on the one APOR GmbH scenario; generalisation to other ROPA contexts is not claimed.
- **Reference-free.** Unlike the NLP metric layer, the judge does not compare against an expert reference document; it scores against the scenario and Art. 30 directly.

D Full Inferential Tables

This appendix consolidates the canonical sample-size derivation and the full inferential statistics for both the survey and the NLP/judge layer. The tables in the Evaluation chapter draw selectively from these; the appendix is reproduced for completeness and reproducibility.

D.1 Sample-Size Derivation

All sample sizes used in the thesis are derived from the raw data files committed to the repository: `python/survey/docs/data.csv` for the survey (one row per participant) and `python/metrics/output/all_scores_combined.csv` for the NLP and judge scores (one row per AI-generated document).

Survey subset	**n**
Recruited participants (raw)	73
Participants with at least one valid session ($p\_0001 \geq 1$)	70
Form responses (mode-level)	69
Ask responses (mode-level)	69
Chat responses (mode-level)	69
Matched-triple participants (valid response in all three modes)	69

Table D.1: Survey sample sizes after consent and quality filtering.

D.2 Survey Inferential Tables

All survey tests use the matched-triple $n = 69$ within-subjects subset. Omnibus is the Friedman test (non-parametric repeated-measures); post-hoc is pairwise Wilcoxon

NLP / judge subset	n
Total documents with parseable NLP scores and judge ratings	113
Form documents	37
Ask documents	37
Chat documents	39
Unique participants represented	47
Participants with at least one document in each mode	28
Strict matched triple (exactly one document per mode)	26

Table D.2: NLP and judge sample sizes. The strict $n = 26$ set is used for all within-subjects tests (Friedman, Wilcoxon signed-rank); the unbalanced $n = 113$ is used for between-groups sensitivity checks.

Survey $\times$ NLP join	n
Joined rows (document with matching survey response)	106
Form joined rows	34
Ask joined rows	34
Chat joined rows	38
Unique participants in the joined sample	44
Participants with survey $\times$ NLP coverage in all three modes	27

Table D.3: Triangulated samples used for confidence–quality analyses. The joined $n = 106$ is the canonical sample for the rank-inversion table; the within-subjects $n = 27$ is used for paired confidence $\times$ quality correlations.

signed-rank with Bonferroni-corrected $\alpha = 0.0167$. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large. For Kendall's W: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

Measure	**n**	χ^2	**df**	**p**	W	**Sig.**
SUS (0–100, higher = better)	69	5.51	2	0.0636	0.040	No
R-TLX composite (1–21)	69	5.24	2	0.0728	0.038	No
R-TLX Mental Demand	69	3.13	2	0.2086	0.023	No
R-TLX Physical Demand	69	2.08	2	0.3535	0.015	No
R-TLX Temporal Demand	69	12.77	2	0.0017	0.093	**Yes**
R-TLX Effort	69	3.78	2	0.1511	0.027	No
R-TLX Frustration	69	8.93	2	0.0115	0.065	**Yes**
R-TLX Performance (positive-coded)	69	2.33	2	0.3114	0.017	No
AI Confidence (item `5_ai`)	69	3.76	2	0.1524	0.027	No
AI Support (6-item composite)	69	1.05	2	0.5919	0.008	No
AI Reuse intention (item `6_ai`)	69	11.12	2	0.0039	0.081	**Yes**

Table D.4: Friedman omnibus across all survey measures ($n = 69$ matched triple).

Mode preference ranking ($n = 69$). The chi-square goodness-of-fit on first-rank votes against a uniform null gives $\chi^2(2, N{=}69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$. The Friedman test on the full rank scores (1–3) gives $\chi^2(2) = 5.07$, $p = 0.0792$, $W = 0.037$. Per-mode rank-1 vote counts: Form 37, Ask 12, Chat 20; mean ranks (lower = better): Form 1.78, Chat 2.07, Ask 2.14.

D.3 NLP and Judge Inferential Tables

NLP and judge tests follow the same family: Friedman omnibus on the strict matched-triple $n = 26$, Wilcoxon signed-rank pairwise post-hoc with Bonferroni-corrected $\alpha = 0.0167$, plus a Kruskal–Wallis sensitivity check on the full unbalanced $n = 113$. Shapiro–Wilk testing identified non-normal distributions in BLEU, BERTScore Recall (Ask, Chat), and SBERT (Form, Ask); non-parametric tests are used throughout.

Measure	**Pair**	**n**	**W**	**p**	**r**	**Sig. (Bonf.)**
SUS	Form vs Ask	60	578.5	0.0131	$+0.368$	**Yes**
SUS	Form vs Chat	58	549.5	0.0177	$+0.358$	No
SUS	Ask vs Chat	61	885.5	0.6660	-0.063	No
R-TLX comp.	Form vs Ask	65	684.0	0.0111	-0.362	**Yes**
R-TLX comp.	Form vs Chat	65	636.0	0.0043	-0.407	**Yes**
R-TLX comp.	Ask vs Chat	63	995.5	0.9318	$+0.012$	No
Temporal Demand	Form vs Ask	43	237.0	0.0038	-0.499	**Yes**
Temporal Demand	Form vs Chat	46	233.0	<0.001	-0.569	**Yes**
Temporal Demand	Ask vs Chat	46	427.0	0.2111	-0.210	No
Frustration	Form vs Ask	52	403.5	0.0091	-0.414	**Yes**
Frustration	Form vs Chat	57	519.5	0.0144	-0.371	**Yes**
Frustration	Ask vs Chat	52	629.0	0.5838	$+0.087$	No
AI Reuse	Form vs Ask	49	402.5	0.0339	$+0.343$	No
AI Reuse	Form vs Chat	43	216.0	0.0015	$+0.543$	**Yes**
AI Reuse	Ask vs Chat	45	454.5	0.4684	$+0.122$	No

Table D.5: Pairwise Wilcoxon signed-rank tests for the survey measures with at least one significant pairwise effect. Sign convention: positive r means the first-named mode scores higher on the measure. Bonferroni-corrected $\alpha = 0.0167$.

Metric	**H**	**p**	**Sig.**	ε^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (mod.)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (mod.)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	<0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (mod.)
SBERT	17.43	<0.001	Yes	0.140 (large)

Table D.6: Kruskal–Wallis sensitivity check on the full unbalanced sample ($n = 113$).

Metric	χ^2	p	Sig.	W
BLEU	6.23	0.044	Yes	0.083 (weak)
ROUGE-1	10.69	0.005	Yes	$\sim$0.18 (mod.)
ROUGE-2	13.46	0.001	Yes	$\sim$0.22 (mod.)
ROUGE-L	11.31	0.004	Yes	$\sim$0.19 (mod.)
METEOR	9.92	0.007	Yes	$\sim$0.17 (mod.)
BERTScore Precision	18.54	$<$0.001	Yes	$\sim$0.30 (strong)
BERTScore Recall	30.77	$<$0.001	Yes	0.410 (strong)
BERTScore F1	13.00	0.002	Yes	$\sim$0.21 (mod.)
SBERT	16.69	$<$0.001	Yes	$\sim$0.27 (mod.)

Table D.7: Friedman omnibus on the strict within-subjects sample ($n = 26$).

Confidence–quality correlations. Spearman ρ between item `5_ai` (AI legal-basis confidence) and document quality, computed within each mode on the joined survey $\times$ NLP sample. Form ($n = 34$): $\rho = -0.22$ ($p = 0.22$) vs. BERTScore F1; $\rho = +0.07$ ($p = 0.70$) vs. Judge Overall. Ask ($n = 34$): $\rho = -0.23$ ($p = 0.19$) vs. BERTScore F1; $\rho = -0.21$ ($p = 0.24$) vs. Judge Overall. Chat ($n = 38$): $\rho = +0.28$ ($p = 0.09$) vs. BERTScore F1; $\rho = +0.41$ ($p = 0.012$) vs. Judge Overall. Pooled ($n = 106$): $\rho = -0.02$ ($p = 0.88$) vs. BERTScore F1; $\rho = +0.14$ ($p = 0.14$) vs. Judge Overall.

Metric	Pair	W	p	r	Sig. (Bonf.)
BLEU	Form vs Ask	108	0.089	$+0.385$	No
BLEU	Form vs Chat	133	0.291	-0.242	No
BLEU	Ask vs Chat	74	0.009	-0.578	**Yes**
ROUGE-1	Form vs Ask	77	0.011	$+0.561$	**Yes**
ROUGE-1	Form vs Chat	147	0.483	-0.162	No
ROUGE-1	Ask vs Chat	63	0.003	-0.641	**Yes**
ROUGE-2	Form vs Ask	30	$<$0.001	$+0.829$	**Yes**
ROUGE-2	Form vs Chat	136	0.328	$+0.225$	No
ROUGE-2	Ask vs Chat	71	0.007	-0.595	**Yes**
ROUGE-L	Form vs Ask	44	$<$0.001	$+0.749$	**Yes**
ROUGE-L	Form vs Chat	161	0.727	$+0.083$	No
ROUGE-L	Ask vs Chat	66	0.004	-0.624	**Yes**
METEOR	Form vs Ask	33	$<$0.001	$+0.812$	**Yes**
METEOR	Form vs Chat	137	0.340	$+0.219$	No
METEOR	Ask vs Chat	84	0.019	-0.521	No
BERTScore Precision	Form vs Ask	63	0.003	$+0.641$	**Yes**
BERTScore Precision	Form vs Chat	106	0.080	-0.396	No
BERTScore Precision	Ask vs Chat	40	$<$0.001	-0.772	**Yes**
BERTScore Recall	Form vs Ask	0	$<$0.001	$+1.000$	**Yes**
BERTScore Recall	Form vs Chat	55	0.001	$+0.687$	**Yes**
BERTScore Recall	Ask vs Chat	99	0.053	-0.436	No
BERTScore F1	Form vs Ask	24	$<$0.001	$+0.863$	**Yes**
BERTScore F1	Form vs Chat	172	0.940	$+0.020$	No
BERTScore F1	Ask vs Chat	59	0.002	-0.664	**Yes**
SBERT	Form vs Ask	110	0.099	$+0.373$	No
SBERT	Form vs Chat	56	0.002	-0.681	**Yes**
SBERT	Ask vs Chat	26	$<$0.001	-0.852	**Yes**

Table D.8: Pairwise Wilcoxon signed-rank tests on the strict within-subjects sample ($n = 26$). Sign convention: positive r means the first-named mode scores higher on the metric. Bonferroni-corrected $\alpha = 0.0167$.

E Inferential Statistics in Full

This appendix reproduces the complete inferential test output referenced in Chapter 4. Sample-size definitions follow Chapter 3: the within-subjects survey tests use $n = 69$, the within-subjects document-quality tests use the matched-triple subset of $n = 26$, and between-groups sensitivity checks on the unbalanced full sample use $n = 113$.

E.1 Survey Inferential Tests

E.1.1 Normality (Shapiro–Wilk, $n = 69$)

Table E.1: Shapiro–Wilk normality tests for survey measures by mode.

Measure	Mode	W	p	Normal?
SUS	Form	0.944	0.004	No
SUS	Ask	0.976	0.210	Yes
SUS	Chat	0.927	0.001	No
R-TLX composite	Form	0.940	0.003	No
R-TLX composite	Ask	0.926	0.001	No
R-TLX composite	Chat	0.945	0.005	No
AI Support (6-item)	Form	0.923	< 0.001	No
AI Support (6-item)	Ask	0.965	0.048	No
AI Support (6-item)	Chat	0.952	0.010	No
AI Confidence (5_{ai})	Form	0.846	< 0.001	No
AI Confidence (5_{ai})	Ask	0.856	< 0.001	No
AI Confidence (5_{ai})	Chat	0.864	< 0.001	No
AI Reuse (6_{ai})	Form	0.799	< 0.001	No
AI Reuse (6_{ai})	Ask	0.872	< 0.001	No
AI Reuse (6_{ai})	Chat	0.861	< 0.001	No

Non-normal distributions are detected for nearly every measure. Non-parametric tests (Friedman omnibus and Wilcoxon signed-rank post-hoc) are used throughout.

E.1.2 Friedman Omnibus ($n = 69$ matched triple)

Table E.2: Friedman omnibus tests across the three modes for survey measures. Kendall's W: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

Measure	χ^2	df	p	W	Significant
SUS (0–100)	5.51	2	0.064	0.040	No
R-TLX composite (1–21)	5.24	2	0.073	0.038	No
R-TLX Mental Demand	3.13	2	0.209	0.023	No
R-TLX Physical Demand	2.08	2	0.354	0.015	No
R-TLX Temporal Demand	12.77	2	0.002	0.093	**Yes**
R-TLX Effort	3.78	2	0.151	0.027	No
R-TLX Frustration	8.93	2	0.012	0.065	**Yes**
R-TLX Performance (pos.)	2.33	2	0.311	0.017	No
AI Confidence (5_{ai})	3.76	2	0.152	0.027	No
AI Support (6-item composite)	1.05	2	0.592	0.008	No
AI Reuse intention (6_{ai})	11.12	2	0.004	0.081	**Yes**

E.1.3 Pairwise Wilcoxon Signed-Rank Post-hoc (Bonferroni $\alpha = 0.0167$)

Post-hoc tests are reported for measures with a significant Friedman result and, for completeness, for SUS and the R-TLX composite. Sign convention: positive r means the first-named mode ranks higher. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

E.1.4 Mode Preference Ranking

A chi-square goodness-of-fit test on the rank-1 vote distribution against a uniform null gives $\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$ (medium effect). The Friedman test on the full rank scores (1–3) is non-significant: $\chi^2(2) = 5.07$, $p = 0.079$, $W = 0.037$.

Table E.3: Pairwise Wilcoxon signed-rank post-hoc tests for survey measures.

Measure	Pair	n	W	p	r	Sig (Bonf)
SUS	Form vs Ask	60	578.5	0.013	$+0.368$ medium	**Yes**
SUS	Form vs Chat	58	549.5	0.018	$+0.358$ medium	No
SUS	Ask vs Chat	61	885.5	0.666	-0.063 negligible	No
R-TLX composite	Form vs Ask	65	684.0	0.011	-0.362 medium	**Yes**
R-TLX composite	Form vs Chat	65	636.0	0.004	-0.407 medium	**Yes**
R-TLX composite	Ask vs Chat	63	995.5	0.932	$+0.012$ negligible	No
Temporal Demand	Form vs Ask	43	237.0	0.004	-0.499 medium	**Yes**
Temporal Demand	Form vs Chat	46	233.0	< 0.001	-0.569 large	**Yes**
Temporal Demand	Ask vs Chat	46	427.0	0.211	-0.210 small	No
Frustration	Form vs Ask	52	403.5	0.009	-0.414 medium	**Yes**
Frustration	Form vs Chat	57	519.5	0.014	-0.371 medium	**Yes**
Frustration	Ask vs Chat	52	629.0	0.584	$+0.087$ negligible	No
AI Reuse (6_{ai})	Form vs Ask	49	402.5	0.034	$+0.343$ medium	No
AI Reuse (6_{ai})	Form vs Chat	43	216.0	0.002	$+0.543$ large	**Yes**
AI Reuse (6_{ai})	Ask vs Chat	45	454.5	0.468	$+0.122$ small	No

E.2 NLP Metric Inferential Tests

E.2.1 Kruskal–Wallis Sensitivity Check ($n = 113$, unbalanced)

Table E.4: Kruskal–Wallis H tests on the unbalanced full document set. Reported as a between-groups sensitivity check, not the primary post-hoc family.

Metric	H	p	Sig ($\alpha = 0.05$)	ε^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (moderate)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (moderate)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	< 0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (moderate)
SBERT (ModernBERT)	17.43	< 0.001	Yes	0.140 (large)

Table E.5: Friedman omnibus tests on the matched-triple subset of $n = 26$ participants. Kendall's W approximate where the metric required tie correction.

Metric	χ^2	p	Sig	W
BLEU	6.23	0.044	Yes	0.083 (weak)
ROUGE-1	10.69	0.005	Yes	$\sim$0.18 (moderate)
ROUGE-2	13.46	0.001	Yes	$\sim$0.22 (moderate)
ROUGE-L	11.31	0.004	Yes	$\sim$0.19 (moderate)
METEOR	9.92	0.007	Yes	$\sim$0.17 (moderate)
BERTScore Precision	18.54	< 0.001	Yes	$\sim$0.30 (strong)
BERTScore Recall	30.77	< 0.001	Yes	0.410 (strong)
BERTScore F1	13.00	0.002	Yes	$\sim$0.21 (moderate)
SBERT (ModernBERT)	16.69	< 0.001	Yes	$\sim$0.27 (moderate)

E.2.2 Friedman Omnibus ($n = 26$ matched triple)

E.2.3 Pairwise Wilcoxon Signed-Rank Post-hoc ($n = 26$, primary)

Bonferroni-corrected $\alpha = 0.0167$. Effect size: matched-pairs rank-biserial r (positive when the first-named mode in the pair scores higher).

E.2.4 Mann–Whitney U Sensitivity Check ($n = 113$, unbalanced)

The Mann–Whitney U test is reported as a between-groups sensitivity check on the unbalanced full sample. Bonferroni-corrected $\alpha = 0.0167$. Only significant comparisons are listed; non-significant comparisons are omitted for brevity.

The sensitivity check broadly agrees with the matched-pairs result on direction (Ask $<$ Form, Ask $<$ Chat on most metrics), with the matched analysis producing larger and more consistent effect sizes — as expected for a within-subjects design.

E.3 Confidence–Quality Correlations

Spearman rank correlation ρ between the primary confidence measure (item 5_{ai}, "I am confident the AI identified the correct legal basis") and document-quality mea-

Table E.6: Pairwise Wilcoxon signed-rank post-hoc tests for document-level NLP metrics on the matched-triple subset.

Metric	Pair	W	p	r	Direction	Sig (Bonf)
BLEU	Form vs Ask	108	0.089	$+0.385$ medium	Form $>$ Ask	No
BLEU	Form vs Chat	133	0.291	-0.242 small	Chat $>$ Form	No
BLEU	Ask vs Chat	74	0.009	-0.578 large	Chat $>$ Ask	**Yes**
ROUGE-1	Form vs Ask	77	0.011	$+0.561$ large	Form $>$ Ask	**Yes**
ROUGE-1	Form vs Chat	147	0.483	-0.162 small	Chat $>$ Form	No
ROUGE-1	Ask vs Chat	63	0.003	-0.641 large	Chat $>$ Ask	**Yes**
ROUGE-2	Form vs Ask	30	< 0.001	$+0.829$ large	Form $>$ Ask	**Yes**
ROUGE-2	Form vs Chat	136	0.328	$+0.225$ small	Form $>$ Chat	No
ROUGE-2	Ask vs Chat	71	0.007	-0.595 large	Chat $>$ Ask	**Yes**
ROUGE-L	Form vs Ask	44	< 0.001	$+0.749$ large	Form $>$ Ask	**Yes**
ROUGE-L	Form vs Chat	161	0.727	$+0.083$ negligible	Form $>$ Chat	No
ROUGE-L	Ask vs Chat	66	0.004	-0.624 large	Chat $>$ Ask	**Yes**
METEOR	Form vs Ask	33	< 0.001	$+0.812$ large	Form $>$ Ask	**Yes**
METEOR	Form vs Chat	137	0.340	$+0.219$ small	Form $>$ Chat	No
METEOR	Ask vs Chat	84	0.019	-0.521 large	Chat $>$ Ask	No (above B
BERTScore Precision	Form vs Ask	63	0.003	$+0.641$ large	Form $>$ Ask	**Yes**
BERTScore Precision	Form vs Chat	106	0.080	-0.396 medium	Chat $>$ Form	No
BERTScore Precision	Ask vs Chat	40	< 0.001	-0.772 large	Chat $>$ Ask	**Yes**
BERTScore Recall	Form vs Ask	0	< 0.001	$+1.000$ large	Form $>$ Ask	**Yes**
BERTScore Recall	Form vs Chat	55	0.001	$+0.687$ large	Form $>$ Chat	**Yes**
BERTScore Recall	Ask vs Chat	99	0.053	-0.436 medium	Chat $>$ Ask	No
BERTScore F1	Form vs Ask	24	< 0.001	$+0.863$ large	Form $>$ Ask	**Yes**
BERTScore F1	Form vs Chat	172	0.940	$+0.020$ negligible	Form $>$ Chat	No
BERTScore F1	Ask vs Chat	59	0.002	-0.664 large	Chat $>$ Ask	**Yes**
SBERT	Form vs Ask	110	0.099	$+0.373$ medium	Form $>$ Ask	No
SBERT	Form vs Chat	56	0.002	-0.681 large	Chat $>$ Form	**Yes**
SBERT	Ask vs Chat	26	< 0.001	-0.852 large	Chat $>$ Ask	**Yes**

Table E.7: Mann–Whitney U between-groups sensitivity check on the unbalanced full sample of $n = 113$ documents.

Metric	Comparison	p	r	Sig
BERTScore Recall	Form $>$ Ask	< 0.001	0.756 (large)	Yes
BERTScore Recall	Form $>$ Chat	< 0.001	0.512 (large)	Yes
BERTScore F1	Form $>$ Ask	0.004	0.386 (medium)	Yes
BERTScore F1	Chat $>$ Ask	0.011	0.340 (medium)	Yes
SBERT	Chat $>$ Form	< 0.001	0.505 (large)	Yes
SBERT	Chat $>$ Ask	< 0.001	0.453 (large)	Yes
ROUGE-2	Form $>$ Ask	0.006	0.369 (medium)	Yes
ROUGE-2	Chat $>$ Ask	0.011	0.342 (medium)	Yes
METEOR	Form $>$ Ask	0.003	0.407 (medium)	Yes

sures, computed on the survey $\times$ NLP joined sample. Per-mode joined samples: Form $n = 34$, Ask $n = 34$, Chat $n = 38$; pooled $n = 106$.

Table E.8: Spearman rank correlations: primary confidence (5_{ai}) versus document-quality measures, by mode.

	5_{ai} vs BERTScore F1			5_{ai} vs Judge Overall		
Mode	n	ρ	p	n	ρ	p
Form	34	-0.22	0.22	34	$+0.07$	0.70
Ask	34	-0.23	0.19	34	-0.21	0.24
Chat	38	$+0.28$	0.09	38	$+0.41$	**0.012**
Pooled	106	-0.02	0.88	106	$+0.14$	0.14

For comparison, the secondary 6-item Perceived AI Support composite produces weaker mode-specific correlations with BERTScore F1 (Form $\rho = -0.22$, Ask $\rho = -0.04$, Chat $\rho = +0.20$) and a small but significant pooled correlation with Judge Overall ($\rho = +0.21$, $p = 0.033$, $n = 106$).

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Declaration

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.

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